

Clustering program

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Chapter 1

Unit Testing Suite - File Index

1.1 Test Source Files

1.1.1 Core Test Programs (7 files) - UPDATED

1. [test_maths.f90](#) (29.4 KB) - EXPANDED

- Tests: 31 tests (updated from 4)
- Modules tested: `maths`
- Features: Cross product, coordinate transformations, angle calculations, RMSD, center of geometry, distance calculations
- Dependencies: [maths.f90](#), [mod_pdb.f90](#)
- Lines: 850

2. [test_matrix.f90](#) (13.1 KB)

- Tests: 8 tests
- Modules tested: `mod_matrix`
- Features: Matrix I/O, array I/O, Z-coordinate calculations, RMSD matrix properties
- Dependencies: [mod_matrix.f90](#), [maths.f90](#), [mod_pdb.f90](#)
- Lines: 370

3. [test_threshold.f90](#) (13.3 KB) - UPDATED

- Tests: 9 tests (updated from 6)
- Modules tested: `mod_threshold`
- Features: Z-coordinate arrays, minimum distance, angle calculations, array sorting
- Dependencies: [mod_threshold.f90](#), [maths.f90](#)
- Lines: 375

4. [test_pdb.f90](#) (7.4 KB)

- Tests: 8 tests
- Modules tested: `mod_pdb`
- Features: PDB structures, atoms, residues, chains
- Dependencies: [mod_pdb.f90](#)
- Lines: 210

5. [test_clustering.f90](#) (9.8 KB)

- Tests: 5 tests
- Modules tested: [mod_clust_algorithm](#)
- Features: Clustering statistics, simple clustering, cluster output writing
- Dependencies: [mod_clust_algorithm.f90](#), [read_input.f90](#), [mod_pdb.f90](#), [mod_assoc.f90](#), [maths.f90](#)
- Lines: 275

6. [test_assoc.f90](#) (13.5 KB) [NEW]

- Tests: 11 tests
- Modules tested: [mod_assoc](#)
- Features: Association file reading, object allocation, file parsing
- Dependencies: [mod_assoc.f90](#)
- Lines: 380

7. [test_read_input.f90](#) (12.6 KB) [NEW]

- Tests: 10 tests
- Modules tested: [read_input](#) (wrapper module)
- Features: Input file reading, structure creation, error handling
- Dependencies: [read_input.f90](#), [mod_pdb.f90](#), [mod_assoc.f90](#)
- Lines: 355

1.2 Documentation Files

1. [TESTING.md](#) (7.2 KB) - UPDATED

- Comprehensive testing guide with all 7 test files
- Compilation instructions for each test
- Updated test coverage table (82 tests total)
- Best practices and troubleshooting
- Instructions for adding new tests

2. [TEST_SUMMARY.md](#) (6.8 KB) - UPDATED

- Overview of entire test suite with 82 tests
- Summary of all 7 test programs
- Test coverage matrix with new modules
- Example output
- Overall statistics table

3. [QUICK_TEST.md](#) (3.2 KB) - UPDATED

- Quick reference for running all 7 test suites
- Command summary table with new tests
- File overview with test counts
- Pro tips

4. [README.md](#) (3.5 KB) - UPDATED

- Test framework overview
- Directory structure with all 7 test programs

- Quick start guide
- Test coverage table (82 tests)
- Build requirements and verification

5. [INDEX.md](#) (This file) - UPDATED

- File listing and descriptions for all 7 test programs
- Documentation index with 5 files

1.3 Automation Files

1. [run_all_tests.sh](#) (2.8 KB) - UPDATED

- Bash script to compile and run all 7 tests
- Automatic dependency tracking for new tests
- Summary reporting for all test programs
- Exit code handling (0 = success, 1 = failure)

1.4 Modified Files

1. **Makefile** - UPDATED

- Added test targets for `test_assoc` and `test_read_input`
- Updated `test` target to run all 7 tests
- Updated `clean_tests` target for new executables
- Integrated with existing build system

1.5 Quick Start

```
cd clustering_program/src/  
# Run all tests  
make test  
# Or use the automated script  
bash run_all_tests.sh  
# Individual test  
make test_math  
make test_pdb
```

1.6 Test Distribution

```
Mathematical Operations (4 tests)  
- cross product  
- coordinate transformations  
Matrix Operations (4 tests)  
- allocation  
- symmetry properties  
Array/Threshold Operations (6 tests)  
- statistics  
- bounds checking  
PDB Data Structures (8 tests)  
- file allocation  
- atom properties  
- residue management  
Clustering Algorithms (6 tests)  
- distance matrix properties  
- clustering helpers
```

1.7 Testing Workflow

1. Build Tests (`make test`)
 - |
 - v
2. Compile each test program
 - Dependency resolution
 - Link with required modules
 - |
 - v
3. Execute tests sequentially
 - Each test reports pass/fail
 - Detailed error messages on failure
 - Summary statistics
 - |
 - v
4. Report results
 - Total tests run
 - Passed count
 - Failed count
 - Exit code (0=success, 1=failure)

1.8 Next Steps

1. **Run tests:** `make test`
2. **Read results:** Check PASS/FAIL output
3. **Add tests:** Follow instructions in [TESTING.md](#)
4. **Integrate CI:** Set up GitHub Actions (optional)
5. **Performance:** Run with profiling tools if needed

1.9 Files Generated by Tests

When tests run, the following executables are created (removed by `make clean_tests`):

- `test_maths`
- `test_matrix`
- `test_threshold`
- `test_pdb`
- `test_clustering`

1.10 Support & References

- [TESTING.md](#) - Detailed guide with troubleshooting
- [TEST_SUMMARY.md](#) - Full overview and future enhancements
- [QUICK_TEST.md](#) - Common commands and tips

Created: February 2026 **Total Tests:** 28 **Coverage:** 5 modules, 11 functions

Chapter 2

Quick Reference: Running Unit Tests

2.1 TL;DR

```
cd clustering_program/src/  
make test          # Compile and run all 82 tests
```

2.2 Test Commands

Command	Description
make test	Compile and run all 82 tests (7 test programs)
make test_maths	Test mathematical operations (31 tests)
make test_matrix	Test matrix operations (8 tests)
make test_threshold	Test threshold arrays (9 tests)
make test_pdb	Test PDB structures (8 tests)
make test_clustering	Test clustering algorithms (5 tests)
make test_assoc	Test association file reading (11 tests) [NEW]
make test_read_input	Test read_input module (10 tests) [NEW]
make clean_tests	Remove test executables
bash run_all_tests.sh	Run all tests with detailed output

2.3 Test Files

File	Tests	Lines	Purpose
test_maths.f90	31	850	Cross product, transformations, RMSD, angles, COG, distance
test_matrix.f90	8	370	Matrix I/O, array I/O, Z-coord, RMSD
test_threshold.f90	9	375	Z-coord, min distance, angles, sorting
test_pdb.f90	8	210	PDB types, atoms, residues
test_clustering.f90	5	275	Statistics, clustering, output writing
test_assoc.f90	11	380	File reading, allocation, parsing [NEW]
test_read_input.f90	10	355	Input wrapper, error handling [NEW]

2.4 Expected Output

```
All tests pass with output like:  
[PASS] Test 1: Test description  
[PASS] Test 2: Another test  
=====  
TEST SUMMARY  
=====  
Total tests:  N
```

```

Passed:      N
Failed:      0
=====

```

2.5 Individual Compilation

```

# Manual compilation (if not using Make)
gfortran -o test_maths test_maths.f90 maths.f90 mod_pdb.f90
./test_maths
gfortran -fopenmp -o test_matrix test_matrix.f90 mod_matrix.f90 maths.f90 mod_pdb.f90
./test_matrix
gfortran -fopenmp -o test_threshold test_threshold.f90
./test_threshold
gfortran -o test_pdb test_pdb.f90 mod_pdb.f90
./test_pdb
gfortran -fopenmp -o test_clustering test_clustering.f90
./test_clustering

```

2.6 Documentation

- [TEST_SUMMARY.md](#) - Overview of test suite
- [TESTING.md](#) - Detailed testing guide

2.7 Key Features

[x] 82 comprehensive tests (updated from 28) [x] 7 test programs (added [test_assoc.f90](#) and [test_read_input.f90](#))
 [x] All major modules covered [x] OpenMP compatible [x] No external data files needed [x] Fast execution (< 2 seconds) [x] Clear pass/fail reporting [x] Easy to extend

Pro Tip: Use `make test 2>&1 | tee test_results.log` to save test output to a file.

Chapter 3

Unit Testing Framework

This directory contains the comprehensive unit testing framework for the clustering program.

3.1 Contents

3.1.1 Test Programs

- [test_maths.f90](#) - Tests for mathematical operations (31 tests)
- [test_matrix.f90](#) - Tests for matrix operations (8 tests)
- [test_threshold.f90](#) - Tests for threshold calculations (9 tests)
- [test_pdb.f90](#) - Tests for PDB file handling (8 tests)
- [test_clustering.f90](#) - Tests for clustering algorithms (5 tests)
- [test_assoc.f90](#) - Tests for association file reading (11 tests) [NEW]
- [test_read_input.f90](#) - Tests for [read_input](#) module (10 tests) [NEW]

3.1.2 Test Automation

- [run_all_tests.sh](#) - Automated script to run all unit tests

3.1.3 Documentation

Document	Purpose	Best For
QUICK_TEST.md	Quick reference guide	Getting started quickly
TESTING.md	Comprehensive testing guide	Understanding all details
TEST_SUMMARY.md	Test suite overview	High-level overview
UNIT_TESTING_COMPLETE.md	Completion summary	Project status
INDEX.md	File index and statistics	Navigation

3.2 Quick Start

3.2.1 Run all tests

```
cd ../src
make test
```

3.2.2 Run specific test

```
cd ../src
make test_maths    # or test_matrix, test_threshold, test_pdb, test_clustering
```

3.2.3 Clean test executables

```
cd ../src
make clean_tests
```

3.3 Where to Start?

1. **New to testing?** - Start with [QUICK_TEST.md](#)
2. **Need details?** - Read [TESTING.md](#)
3. **Want overview?** - See [TEST_SUMMARY.md](#)
4. **Need file list?** - Check [INDEX.md](#)
5. **Project status?** - Read [UNIT_TESTING_COMPLETE.md](#)

3.4 Test Coverage

Module	Tests	Status
maths.f90	31	Yes
mod_matrix.f90	8	Yes
mod_threshold.f90	9	Yes
mod_pdb.f90	8	Yes
mod_clust_algorithm.f90	5	Yes
mod_assoc.f90	11	Yes [NEW]
read_input.f90	10	Yes [NEW]
Total	82	Yes

3.5 Build Requirements

- gfortran compiler
- LAPACK and BLAS libraries
- OpenMP support

3.6 Verification

All 82 unit tests are defined and ready to run. The framework provides:

- Comprehensive test coverage across 7 test files
- Assertion helpers for test validation
- Pass/fail reporting with detailed output
- Error message display
- Command-line integration via Makefile
- Standalone bash script runner

3.7 Notable Behavior

Array Parameter in Clustering:

- The `-array` option is **mandatory** for all clustering methods **except** **RMSD**

- When using `-matrix_type rmsd`, the array parameter is **optional**
- When an array is provided, complexes are sorted by the array values
- If array is not provided for non-RMSD methods, an error is raised
- Without an array, cluster averages and standard deviations are computed from the matrix

Test File Coverage Updates (February 2026):

- Added comprehensive tests for [mod_assoc](#) module (association file reading)
- Added comprehensive tests for [read_input](#) module (input file processing)
- Expanded maths module tests from 2 to 31 tests with detailed coordinate transformation coverage
- Updated all test documentation to reflect current test count (82 total tests)

For detailed information, see [TESTING.md](#) and [UNIT_TESTING_COMPLETE.md](#).

Chapter 4

Unit Testing Suite for Clustering Program

4.1 Summary

A comprehensive unit testing framework has been created for the Fortran clustering program with **7 test programs** covering **82 individual tests** across all major modules.

4.2 Test Programs Created

4.2.1 1. **test_maths.f90** (31 tests)

- Cross product calculation (basis vectors, anti-commutativity, self-product, arbitrary vectors) - 4 tests
- Coordinate transformations (identity, translation, rotation, combinations) - 7 tests
- Angle calculations in 3D and 2D - 8 tests
- RMSD calculations - 3 tests
- Center of geometry computations - 5 tests
- Distance calculations - 4 tests
- Tests fundamental mathematical operations used throughout the program

4.2.2 2. **test_matrix.f90** (8 tests)

- Matrix file I/O operations (write/read matrix) - 2 tests
- Array file I/O operations (write/read array) - 2 tests
- Z-coordinate matrix calculations - 2 tests
- RMSD matrix calculations with symmetry and diagonal verification - 2 tests
- Tests core matrix operations for encounter data

4.2.3 3. **test_threshold.f90** (9 tests)

- Z-coordinate array calculations - 2 tests
- Minimum distance computations - 2 tests
- Angle calculations across arrays - 2 tests
- Array sorting with index tracking - 3 tests
- Tests threshold computation infrastructure

4.2.4 4. **test_pdb.f90** (8 tests)

- PDB structure initialization and allocation - 3 tests
- Atom properties (coordinates, names, residue info, occupancy, B-factors) - 3 tests
- Residue management and chain handling - 2 tests
- Tests PDB data structure integrity

4.2.5 5. **test_clustering.f90** (5 tests)

- Clustering statistics (mean distance, standard deviation) - 2 tests
- Simple clustering algorithm with tight cluster detection - 2 tests
- Cluster output file writing and validation - 1 test
- Tests core clustering algorithm components

4.2.6 6. **test_assoc.f90** (11 tests) [NEW]

- Association/complexes file reading - 6 tests
- Association object allocation - 3 tests
- File size and line counting - 2 tests
- Tests association file I/O and data structures

4.2.7 7. **test_read_input.f90** (10 tests) [NEW]

- Read input wrapper functions - 7 tests
- Error handling and edge cases - 3 tests
- Tests input file processing module

4.3 Files Added

```
clustering_program/unit_tests/
- test_maths.f90          # Mathematical operations tests (31 tests)
- test_matrix.f90        # Matrix operations tests (8 tests)
- test_threshold.f90      # Threshold array tests (9 tests)
- test_pdb.f90           # PDB structure tests (8 tests)
- test_clustering.f90    # Clustering algorithm tests (5 tests)
- test_assoc.f90         # Association file tests (11 tests) [NEW]
- test_read_input.f90    # Read input module tests (10 tests) [NEW]
- run_all_tests.sh       # Automated test runner script
- TESTING.md             # Comprehensive testing documentation
- README.md              # Test suite overview
- TEST_SUMMARY.md        # This file
- ... (other documentation files)
```

4.4 Building Tests

4.4.1 Individual tests:

```
cd clustering_program/src/
# Test specific module
gfortran -fopenmp -o test_maths ../unit_tests/test_maths.f90 maths.o mod_pdb.o
gfortran -fopenmp -o test_matrix ../unit_tests/test_matrix.f90 mod_matrix.o maths.o mod_pdb.o
gfortran -fopenmp -o test_threshold ../unit_tests/test_threshold.f90 mod_threshold.o maths.o
gfortran -o test_pdb ../unit_tests/test_pdb.f90 mod_pdb.o
gfortran -fopenmp -o test_clustering ../unit_tests/test_clustering.f90 mod_clust_algorithm.o read_input.o
mod_pdb.o mod_assoc.o maths.o
gfortran -o test_assoc ../unit_tests/test_assoc.f90 mod_assoc.o
gfortran -o test_read_input ../unit_tests/test_read_input.f90 read_input.o mod_pdb.o mod_assoc.o
```

4.4.2 Run all tests via Make:

```
cd clustering_program/src/
make test          # Compile and run all 7 test suites
make test_maths    # Run specific test
make test_matrix
make test_threshold
make test_pdb
make test_clustering
make test_assoc    # NEW
make test_read_input # NEW
make clean_tests   # Remove test executables only
```

4.4.3 Automated test runner:

```
cd clustering_program/unit_tests/
bash run_all_tests.sh
```

4.5 Test Coverage Matrix

Module	Function	Tests	Type
maths	cross	4	Unit
maths	update_complex	2	Unit
mod_matrix	allocation	2	Unit
mod_matrix	symmetry	2	Unit
mod_threshold	statistics	3	Unit
mod_threshold	bounds	3	Unit
mod_pdb	allocation	3	Unit
mod_pdb	atoms	3	Unit
mod_pdb	residues	2	Unit
mod_clustering	distances	3	Unit
mod_clustering	helpers	3	Unit

Total: 28 unit tests

4.6 Test Features

[x] **Comprehensive coverage** - Tests mathematical operations, data structures, and algorithms [x] **Floating-point tolerance** - Uses appropriate precision (1d-10 to 1d-8) for double precision [x] **Edge case handling** - Tests boundary conditions and special cases [x] **Clear reporting** - Pass/fail output with detailed diagnostics [x] **Automated execution** - Makefile targets and shell script runner [x] **OpenMP compatible** - Handles parallel compilation flags [x] **Modular design** - Independent test programs, easy to extend

4.7 Example Test Output

```
=====
UNIT TESTS FOR MATHS MODULE
=====
Testing cross product...
[PASS] Test 1: i x j = k
[PASS] Test 2: j x i = -k
[PASS] Test 3: v x v = 0
[PASS] Test 4: Arbitrary vectors
Testing coordinate transformation (update_complex)...
[PASS] Test 1: Identity transformation
[PASS] Test 2: Pure translation
=====
TEST SUMMARY
=====
Total tests: 6
Passed:      6
Failed:      0
=====
```

4.8 Example Output

```
=====
UNIT TESTS FOR MATHS MODULE
=====
Testing cross product...
[PASS] Test 1: i x j = k
[PASS] Test 2: j x i = -k
[PASS] Test 3: v x v = 0
[PASS] Test 4: Arbitrary vectors
Testing coordinate transformation (update_complex)...
[PASS] Test 1: Identity transformation
[PASS] Test 2: Pure translation
[PASS] Test 3: XC1-only transformation
[PASS] Test 4: XC2-only transformation
[PASS] Test 5: Rotation-only transformation
[PASS] Test 6: Combined transformations
[PASS] Test 7: Matrix operation verification
... (31 total tests in test_maths.f90)
=====
TEST SUMMARY FOR MATHS
=====
Total tests: 31
Passed:      31
Failed:      0
=====
```

4.9 Overall Statistics

Test File	Tests	Lines	Status
test_maths.f90	31	850	YES
test_matrix.f90	8	370	YES
test_threshold.f90	9	375	YES
test_pdb.f90	8	210	YES
test_clustering.f90	5	275	YES
test_assoc.f90	11	380	YES [NEW]
test_read_input.f90	10	355	YES [NEW]
TOTAL	82	2,815	YES

4.10 Running Tests

4.10.1 Quick test:

```
cd clustering_program/src/
make test          # Runs all 82 tests across 7 test programs
```

4.10.2 Run specific test:

```
cd clustering_program/src/
make test_maths    # Or test_matrix, test_threshold, test_pdb, etc.
make test_assoc    # Run new association tests
make test_read_input # Run new read_input tests
```

4.10.3 Verbose output:

```
cd clustering_program/unit_tests/
bash run_all_tests.sh
```

4.10.4 Individual test debugging:

```
cd clustering_program/src/
gfortran -fopenmp -o test_matrix ../unit_tests/test_matrix.f90 mod_matrix.o maths.o mod_pdb.o
./test_matrix
```

4.11 Documentation

See [TESTING.md](#) for:

- Detailed test descriptions
- Compilation instructions
- Troubleshooting guide
- Instructions for adding new tests
- Best practices and references

4.12 Future Enhancements

- [] Integration tests for multi-module workflows
- [] Performance benchmarks
- [] Code coverage analysis
- [] Continuous integration (GitHub Actions)
- [] Memory leak detection (valgrind)
- [] Additional clustering algorithm tests

4.13 Dependencies

- gfortran (or other Fortran compiler)
- OpenMP (for parallel tests)
- LAPACK/BLAS (existing project dependency)

4.14 Notes

- Tests use standard IEEE 754 double precision (kind=8)
- All tests are deterministic and reproducible
- Tests do not require external data files
- Clean exit codes: 0 (success), 1 (failure)

Created: February 2026

Author: Abraham Muniz-Chicharro

Version: 1.0

Chapter 5

Unit Testing Guide for Clustering Program

5.1 Overview

This directory contains comprehensive unit tests for the Fortran subroutines in the clustering program. The tests verify correctness of mathematical operations, data structures, and algorithmic components.

5.2 Test Files

5.2.1 1. **test_maths.f90**

Tests mathematical operations in the `maths` module:

- **Cross product calculation** (`cross` subroutine) - 4 tests
 - Standard basis vector products ($i \times j = k$)
 - Anti-commutativity ($v \times w = -w \times v$)
 - Self-product is zero ($v \times v = 0$)
 - Arbitrary vector products
- **Coordinate transformations** (`update_complex` subroutine) - 7 tests
 - Identity transformations
 - Pure translation
 - XC1-only transformation
 - XC2-only transformation
 - Rotation-only transformation
 - Combined transformations
 - Verification of matrix operations
- **Angle calculations** (`vectors_angle_3D`, `vectors_angle_2D`) - 8 tests
 - 3D angle calculations with various vector combinations
 - 2D angle calculations with perpendicular and parallel vectors
- **RMSD calculations** (`rmsd` subroutine) - 3 tests
 - Identical structures (zero RMSD)
 - Translated structures
 - Rotated structures
- **Center of geometry** (`calculate_cog` subroutine) - 5 tests
 - Simple geometries

- Symmetric point clouds
- Offset coordinates
- **Distance calculations** (`calculate_distance` subroutine) - 4 tests
 - Unit vectors
 - Arbitrary vectors
 - Zero distance cases

Total: 31 tests

5.2.2 2. **test_matrix.f90**

Tests matrix operations in the `mod_matrix` module:

- **Matrix file I/O** (`write_matrix`, `read_matrix`) - 2 tests
 - Small matrices (5 x 5)
 - Symmetric matrix properties
- **Array file I/O** (`write_array`, `read_array`) - 2 tests
 - Single and double precision arrays
 - Large array serialization
- **Z-coordinate matrix calculation** (`matrix_z_coord`) - 2 tests
 - Identity transformation (coordinates unchanged)
 - Translation transformations
- **RMSD matrix calculation** (`matrix_rmsd_calc`) - 2 tests
 - Symmetry verification
 - Diagonal properties (zeros on diagonal)

Total: 8 tests

5.2.3 3. **test_threshold.f90**

Tests threshold array operations in the `mod_threshold` module:

- **Z-coordinate array calculation** (`array_z_coord`) - 2 tests
 - Identity transformation
 - Translation effects
- **Minimum distance calculation** (`array_atoms_dist`) - 2 tests
 - Distance computations
 - Minimum value detection
- **Angle calculations** (`array_angle`) - 2 tests
 - 3D angle calculations across arrays
 - Multiple vector pair angles
- **Array sorting** (`sort_array`) - 3 tests
 - Ascending order sorting
 - Index tracking
 - Large array handling

Total: 9 tests

5.2.4 4. **test_pdb.f90**

Tests PDB file structures and operations in the `mod_pdb` module:

- **PDB type initialization** - 3 tests
 - File structure allocation
 - Atom array creation
 - Residue array creation
- **Atom properties** - 3 tests
 - Coordinate assignment
 - Name and residue information
 - Occupancy and B-factor values
- **Residue management** - 2 tests
 - Residue property assignment
 - Atom array allocation within residues

Total: 8 tests

5.2.5 5. **test_clustering.f90**

Tests clustering algorithms in the `mod_clust_algorithm` module:

- **Clustering statistics** (`calculate_mean_distance`, `calculate_std_deviation`) - 2 tests
 - Mean distance calculations from clusters
 - Standard deviation computations
- **Simple clustering algorithm** (`simple_clustering`) - 2 tests
 - Tight cluster pair detection
 - Identical point handling
- **Cluster output** (`write_clusters`) - 1 test
 - File output verification
 - Cluster format validation

Total: 5 tests

5.2.6 6. **test_assoc.f90** (NEW)

Tests association/complexes file reading in the `mod_assoc` module:

- **File reading** (`read_assoc_file`) - 6 tests
 - Successfully read complexes from files
 - Correct structure allocation
 - Data verification (natom, residues, chains)
 - Header line parsing
 - Comment line handling
 - Sequential file reading
- **Object allocation** (`allocate_assoc_object`) - 3 tests
 - Memory allocation
 - Array initialization

- Size verification
- **File parsing** (`size_assoc`) - 2 tests
 - Line counting with comment handling
 - Empty file handling

Total: 11 tests

5.2.7 7. **test_read_input.f90** (NEW)

Tests `read_input` module wrapper functions:

- **Association file reading** (`read_input_assoc`) - 7 tests
 - File reading and structure creation
 - Data verification (`natom`, `residues`, `chains`)
 - Header and comment handling
 - Sequential file operations
 - Memory allocation
 - File existence checking
 - Data integrity validation
- **Error handling** - 3 tests
 - Non-existent file handling
 - Invalid file format detection
 - Sequential read verification

Total: 10 tests

- Occupancy and B-factor values

Residue management

- Residue property assignment
- Atom array allocation within residues

5.3 Compilation

5.3.1 Compile individual tests:

```
# Test mathematical operations (requires maths.f90 and mod_pdb.f90)
gfortran -fopenmp -o test_maths test_maths.f90 ../src/maths.o ../src/mod_pdb.o
# Test matrix operations (requires mod_matrix.f90, maths.f90, mod_pdb.f90)
gfortran -fopenmp -o test_matrix test_matrix.f90 ../src/mod_matrix.o ../src/maths.o ../src/mod_pdb.o
# Test threshold operations (requires mod_threshold.f90, maths.f90)
gfortran -fopenmp -o test_threshold test_threshold.f90 ../src/mod_threshold.o ../src/maths.o
# Test PDB operations (requires mod_pdb.f90)
gfortran -o test_pdb test_pdb.f90 ../src/mod_pdb.o
# Test clustering operations (requires mod_clust_algorithm.f90, read_input.f90, etc.)
gfortran -fopenmp -o test_clustering test_clustering.f90 ../src/mod_clust_algorithm.o ../src/read_input.o
../src/mod_pdb.o ../src/mod_assoc.o ../src/maths.o
# Test association file reading (requires mod_assoc.f90)
gfortran -o test_assoc test_assoc.f90 ../src/mod_assoc.o
# Test read_input module (requires read_input.f90, mod_pdb.f90, mod_assoc.f90)
gfortran -o test_read_input test_read_input.f90 ../src/read_input.o ../src/mod_pdb.o ../src/mod_assoc.o
```

5.3.2 Compile all tests at once:

```
# Using make (from src directory)
make test
# Or using the test runner script (from unit_tests directory)
bash run_all_tests.sh
```

5.4 Running Tests

5.4.1 Run individual tests:

```
./test_maths
./test_matrix
./test_threshold
./test_pdb
```

5.4.2 Run all tests:

```
# Create a simple test runner script
bash run_all_tests.sh
```

5.5 Example Output

```
=====
UNIT TESTS FOR MATHS MODULE
=====
Testing cross product...
[PASS] Test 1: i x j = k
[PASS] Test 2: j x i = -k
[PASS] Test 3: v x v = 0
[PASS] Test 4: Arbitrary vectors
Testing coordinate transformation (update_complex)...
[PASS] Test 1: Identity transformation
[PASS] Test 2: Pure translation
=====
TEST SUMMARY
=====
Total tests: 6
Passed:      6
Failed:      0
=====
```

5.6 Test Coverage

Module	Subroutine	Tests	Status
maths	cross	4	Yes
maths	update_complex	7	Yes
maths	vectors_angle_3D	4	Yes
maths	vectors_angle_2D	4	Yes
maths	rmsd	3	Yes
maths	calculate_cog	5	Yes
maths	calculate_distance	4	Yes
mod_matrix	write_matrix / read_matrix	2	Yes
mod_matrix	write_array / read_array	2	Yes
mod_matrix	matrix_z_coord	2	Yes
mod_matrix	matrix_rmsd_calc	2	Yes
mod_threshold	array_z_coord	2	Yes
mod_threshold	array_atoms_dist	2	Yes
mod_threshold	array_angle	2	Yes
mod_threshold	sort_array	3	Yes
mod_pdb	PDB allocation	3	Yes
mod_pdb	atom properties	3	Yes
mod_pdb	residue management	2	Yes
mod_clust_algorithm	clustering statistics	2	Yes
mod_clust_algorithm	simple_clustering	2	Yes
mod_clust_algorithm	write_clusters	1	Yes
mod_assoc	read_assoc_file	6	Yes
mod_assoc	allocate_assoc_object	3	Yes

Module	Subroutine	Tests	Status
mod_assoc	size_assoc	2	Yes
read_input	read_input_assoc	7	Yes
read_input	error_handling	3	Yes

Total: 82 tests

5.7 Adding New Tests

To add tests for additional subroutines:

1. **Create a new test program** following the template:

```

program test_my_module
  implicit none
  integer :: num_tests, num_passed, num_failed

  num_tests = 0
  num_passed = 0
  num_failed = 0

  ! Call test subroutines
  call test_my_subroutine(num_tests, num_passed, num_failed)

  ! Print summary
  if (num_failed > 0) stop 1
end program

```

2. **Implement test subroutines** with clear assertions:

```

subroutine test_my_subroutine(num_tests, num_passed, num_failed)
  ! Test implementation
end subroutine

```

3. **Add compilation rule** to Makefile
4. **Document test** in this README

5.8 Best Practices

1. **Test isolation:** Each test should be independent
2. **Clear naming:** Use descriptive test names
3. **Tolerance handling:** Use appropriate floating-point tolerances (typically 1d-10 or 1d-8)
4. **Edge cases:** Test boundary conditions and special cases
5. **Performance:** Consider testing with realistic data sizes
6. **Documentation:** Include test purpose in comments

5.9 Troubleshooting

5.9.1 Compilation errors:

- Ensure all required modules are compiled first
- Check for missing `-fopenmp` flag if using OpenMP
- Verify Fortran standard compatibility

5.9.2 Runtime failures:

- Check for NaN/Inf values in calculations
- Verify array bounds and allocation
- Review floating-point tolerance settings

5.9.3 Performance issues:

- Monitor memory usage for large matrices
- Use OpenMP pragmas for parallelized loops
- Profile with `gprof` or similar tools

5.10 References

- [Fortran Standard Library](#)
- [OpenMP Fortran API](#)
- [PDB File Format](#)

Author: Abraham Muniz-Chicharro

Version: 1.0

Date: February 2026

Chapter 6

Unit Testing Suite - Complete Overview

6.1 Deliverables

A comprehensive unit testing framework has been successfully created for the clustering program with full documentation and automation.

6.2 Summary Statistics

Item	Count	Details
Test Programs	7	test_maths, test_matrix, test_threshold, test_pdb, test_clustering, test_↔ assoc, test_read_input
Total Tests	82	Distributed across 7 modules (updated from 28)
Source Lines	~2,815	Fortran 90 test code (updated from ~1,100)
Documentation	5 files	TESTING.md , TEST_SUMMARY.md , QUICK_TEST.md , INDEX.md , README.md
Automation Scripts	1	run_all_tests.sh
Modules Tested	7	maths, mod_matrix , mod_threshold , mod_pdb , mod_clust_algorithm , mod_assoc , read_input
Functions Tested	26	cross, update_complex, vectors_angle_3D, vectors_angle_2D, rmsd, calculate_cog, calculate_distance, + 19 others

6.3 Files Created

6.3.1 Test Programs (7 files)

```
unit_tests/  
- test_maths.f90           [31 tests] - Mathematical operations  
- test_matrix.f90         [8 tests]  - Matrix operations  
- test_threshold.f90      [9 tests]  - Threshold arrays  
- test_pdb.f90            [8 tests]  - PDB structures  
- test_clustering.f90     [5 tests]  - Clustering algorithms  
- test_assoc.f90          [11 tests] - Association file reading [NEW]  
- test_read_input.f90     [10 tests] - Read input module [NEW]
```

6.3.2 Documentation (5 files)

```
unit_tests/  
- TESTING.md               - Comprehensive testing guide (7.2 KB)  
- TEST_SUMMARY.md         - Test suite overview (6.8 KB)  
- QUICK_TEST.md           - Quick reference guide (3.2 KB)  
- INDEX.md                - File index and statistics (4.0 KB)  
- README.md               - Unit testing framework overview (3.5 KB)
```

6.3.3 Automation (1 file)

```
unit_tests/
```

```
- run_all_tests.sh          - Automated test runner (executable) - UPDATED
```

6.3.4 Modified Files (1 file)

```
src/  
- Makefile                  - Added test_assoc and test_read_input targets
```

6.4 Test Coverage Breakdown

6.4.1 Mathematical Operations (31 tests)

- Cross product: 4 tests (basis vectors, anti-commutativity, self-product, arbitrary)
- Coordinate transformations: 7 tests (identity, translation, rotations, combinations)
- Angle calculations: 8 tests (3D and 2D angle computations)
- RMSD calculations: 3 tests (identity, translation, rotation structures)
- Center of geometry: 5 tests (simple, symmetric, offset geometries)
- Distance calculations: 4 tests (unit vectors, arbitrary vectors, zero distance)

6.4.2 Matrix Operations (8 tests)

- [x] Matrix I/O: 2 tests (write/read matrices)
- [x] Array I/O: 2 tests (write/read arrays)
- [x] Z-coordinate matrix: 2 tests (identity, translation)
- [x] RMSD matrix: 2 tests (symmetry, diagonal properties)

6.4.3 Threshold Array Operations (9 tests)

- [x] Z-coordinate array: 2 tests (identity, translation)
- [x] Minimum distance: 2 tests (distance computation, min detection)
- [x] Angle calculations: 2 tests (3D angles, multiple pairs)
- [x] Array sorting: 3 tests (ascending order, index tracking, large arrays)

6.4.4 PDB Data Structures (8 tests)

- [x] Allocation: 3 tests (atoms, residues, chains)
- [x] Atom properties: 3 tests (coordinates, names, residue info, occupancy)
- [x] Residue management: 2 tests (properties, atom allocation)

6.4.5 Clustering Algorithms (5 tests)

- [x] Clustering statistics: 2 tests (mean distance, standard deviation)
- [x] Simple clustering: 2 tests (tight cluster detection, identical points)
- [x] Cluster output: 1 test (file writing, format validation)

6.4.6 Association File Reading (11 tests) [NEW]

- [x] File reading: 6 tests (read complexes, structure allocation, data verification, header/comment handling, sequential reading)
- [x] Object allocation: 3 tests (memory allocation, initialization, size verification)
- [x] File parsing: 2 tests (line counting, comment handling)

6.4.7 Read Input Module (10 tests) [NEW]

- [x] Input wrapper functions: 7 tests (file reading, structure creation, data verification, sequential operations, memory allocation, file existence, data integrity)
- [x] Error handling: 3 tests (non-existent files, invalid formats, sequential read verification)

6.5 Quick Start

6.5.1 Run all tests:

```
cd clustering_program/src/
make test
```

6.5.2 Run specific test:

```
make test_maths
make test_pdb
```

6.5.3 Use automation script:

```
bash run_all_tests.sh
```

6.5.4 Manual compilation:

```
gfortran -o test_maths test_maths.f90 maths.f90 mod_pdb.f90
./test_maths
```

6.6 Documentation Overview

Document	Purpose	Key Sections
TESTING.md	Comprehensive guide	Compilation, coverage table, best practices, troubleshooting
TEST_SUMMARY.md	Suite overview	Summary, file descriptions, coverage matrix, future enhancements
QUICK_TEST.md	Quick reference	Commands, file overview, pro tips
INDEX.md	File index	Statistics, workflows, support references

6.7 Make Targets

```
make test           # Compile and run all tests
make test_maths     # Run maths tests
make test_matrix     # Run matrix tests
make test_threshold  # Run threshold tests
make test_pdb        # Run PDB tests
make test_clustering # Run clustering tests
make clean_tests     # Remove test executables
```

6.8 Key Features

[x] **Comprehensive** - 28 tests covering all major modules [x] **Automated** - Single command to run all tests [x] **Well-documented** - 4 documentation files with examples [x] **Easy to extend** - Clear structure for adding new tests [x]

Fast - All tests complete in under 1 second [x] **Standalone** - No external data files required [x] **OpenMP compatible**
 - Handles parallel compilation [x] **Exit codes** - Proper error handling (0=success, 1=failure)

6.9 Test Execution Flow

```
User runs: make test
|
v
Makefile triggers test targets in order:
- test_maths      -> 4 tests -> PASS/FAIL
- test_matrix     -> 4 tests -> PASS/FAIL
- test_threshold  -> 6 tests -> PASS/FAIL
- test_pdb        -> 8 tests -> PASS/FAIL
- test_clustering -> 6 tests -> PASS/FAIL
|
v
Final summary:
- Total: 28 tests
- Passed: 28
- Failed: 0
- Exit code: 0 (success)
```

6.10 Next Steps

1. **Verify setup:** `make test` in `clustering_program/src/`
2. **Review results:** Check output for PASS/FAIL status
3. **Read documentation:** Start with [QUICK_TEST.md](#)
4. **Add new tests:** Follow template in [TESTING.md](#)
5. **Integrate CI:** Set up GitHub Actions for automated testing (optional)

6.11 Example Test Output

```
=====
UNIT TESTS FOR MATHS MODULE
=====
Testing cross product...
[PASS] Test 1: i x j = k
[PASS] Test 2: j x i = -k
[PASS] Test 3: v x v = 0
[PASS] Test 4: Arbitrary vectors
Testing coordinate transformation (update_complex)...
[PASS] Test 1: Identity transformation
[PASS] Test 2: Pure translation
=====
TEST SUMMARY
=====
Total tests: 6
Passed:      6
Failed:      0
=====
```

6.12 Documentation Location

All files are in: `/home/abraham/cellulose_project/clustering_program/src/`

- Start with: [QUICK_TEST.md](#) (quick reference)
- Details: [TESTING.md](#) (comprehensive guide)
- Overview: [TEST_SUMMARY.md](#) (full suite description)
- Index: [INDEX.md](#) (file organization)

6.13 Verification Checklist

- [x] 5 test programs created with 28 tests
- [x] All major modules covered (5 modules)
- [x] Comprehensive documentation (4 files)
- [x] Automated test runner (shell script)
- [x] Makefile integration with test targets
- [x] Clear pass/fail reporting
- [x] Proper error handling and exit codes
- [x] No external dependencies (only uses gfortran)
- [x] Examples and troubleshooting guides
- [x] Instructions for extending tests

6.14 Best Practices Implemented

[x] **Modularity** - Each module has dedicated test file [x] **Clarity** - Descriptive test names and output [x] **Isolation** - Tests are independent [x] **Tolerance** - Appropriate floating-point precision (1d-10 to 1d-8) [x] **Edge cases** - Boundary conditions tested [x] **Documentation** - Well-commented test code [x] **Automation** - Make targets and shell scripts [x] **Reporting** - Detailed pass/fail with diagnostics

6.15 Support

If tests fail:

1. Check [TESTING.md](#) troubleshooting section
2. Verify gfortran is installed: `gfortran --version`
3. Ensure OpenMP is available: `gfortran -fopenmp --version`
4. Review test output for specific error messages
5. Compile individual tests for debugging

6.16 Summary

A production-ready unit testing suite with:

- [x] 28 comprehensive tests
- [x] 5 independent test programs
- [x] Full documentation with examples
- [x] Automated execution via Make
- [x] Clear reporting and diagnostics
- [x] Easy extensibility

Ready to use! Run `cd clustering_program/src/ && make test`

Date Created: February 6, 2026

Test Suite Version: 1.0

Author: Abraham Muniz-Chicharro

Status: [x] Complete and Ready for Use

Chapter 7

Modules Index

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9.1 File List

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Chapter 10

Module Documentation

10.1 maths Module Reference

Mathematical utilities and coordinate transformation routines.

Functions/Subroutines

- subroutine [cross](#) (v1, v2, v3)
Compute the cross product of two 3D vectors.
- subroutine [update_complex](#) (xc1, xc2, trans, rot_vx, rot_vy, nb_atoms, coords, new_coords)
Apply translation and rotation to a set of atomic coordinates.
- subroutine [vectors_angle_3d](#) (vector1, vector2, theta_degrees)
Compute the angle in degrees between two 3D vectors.
- subroutine [vectors_angle_2d](#) (vector1, vector2, theta_degrees)
Compute the angle in degrees between two vectors projected to 2D.
- subroutine [rmsd](#) (rmsd_value, nb_atoms, coords1, coords2)
Compute the root-mean-square deviation (RMSD) between two sets of coordinates.
- subroutine [calculate_cog](#) (cog, coords, nb_atoms)
Compute the center of geometry (centroid) of a set of points.
- subroutine [calculate_distance](#) (p1, p2, dist)
Compute Euclidean distance between two 3D points.

10.1.1 Detailed Description

Mathematical utilities and coordinate transformation routines.

This module provides fundamental math operations including cross products, coordinate transformations, angle calculations, and rotation matrices.

Author

Abraham Muñiz-Chicharro

10.1.2 Function/Subroutine Documentation

10.1.2.1 [calculate_cog\(\)](#)

```
subroutine maths::calculate_cog (  
    real (kind=8), dimension(3), intent(out) cog,  
    real (kind=8), dimension(nb_atoms, 3), intent(in) coords,  
    integer, intent(in) nb_atoms )
```

Compute the center of geometry (centroid) of a set of points.

Sums the coordinates and divides by the number of points to produce the centroid.

Parameters

out	<i>cog</i>	Centroid vector (3 elements)
in	<i>coords</i>	Input coordinates, shape (nb_atoms,3)
in	<i>nb_atoms</i>	Number of atoms / points

10.1.2.2 calculate_distance()

```
subroutine maths::calculate_distance (
    real (kind=8), dimension(3), intent(in) p1,
    real (kind=8), dimension(3), intent(in) p2,
    real (kind=8), intent(out) dist )
```

Compute Euclidean distance between two 3D points.

Parameters

in	<i>p1</i>	First point (3 elements)
in	<i>p2</i>	Second point (3 elements)
out	<i>dist</i>	Euclidean distance $\ p1 - p2\ $

10.1.2.3 cross()

```
subroutine maths::cross (
    real ( kind=8 ), dimension ( 3 ), intent(in) v1,
    real ( kind=8 ), dimension ( 3 ), intent(in) v2,
    real ( kind=8 ), dimension ( 3 ), intent(out) v3 )
```

Compute the cross product of two 3D vectors.

Computes $v3 = v1 \times v2$ using the standard right-hand rule.

Parameters

in	<i>v1</i>	Left-hand operand vector (3 elements)
in	<i>v2</i>	Right-hand operand vector (3 elements)
out	<i>v3</i>	Resulting cross product (3 elements)

10.1.2.4 rmsd()

```
subroutine maths::rmsd (
    real(kind=8), intent(out) rmsd_value,
    integer, intent(in) nb_atoms,
    real(kind=8), dimension(:, :), intent(in) coords1,
    real(kind=8), dimension(:, :), intent(in) coords2 )
```

Compute the root-mean-square deviation (RMSD) between two sets of coordinates.

Calculates $RMSD = \sqrt{(1/N) * \sum_i \|coords1(i,:) - coords2(i,:)\|^2}$.

Parameters

out	<i>rmsd_value</i>	Resulting RMSD (scalar)
in	<i>nb_atoms</i>	Number of atoms / points (integer)
in	<i>coords1</i>	First coordinate set, shape (nb_atoms,3)

Parameters

in	<i>coords2</i>	Second coordinate set, shape (nb_atoms,3)
----	----------------	---

10.1.2.5 update_complex()

```

subroutine maths::update_complex (
    real ( kind = 8 ), dimension ( 3 ), intent(in) xc1,
    real ( kind = 8 ), dimension ( 3 ), intent(in) xc2,
    real ( kind = 8 ), dimension ( 3 ), intent(in) trans,
    real ( kind = 8 ), dimension ( 3 ), intent(in) rot_vx,
    real ( kind = 8 ), dimension ( 3 ), intent(in) rot_vy,
    integer nb_atoms,
    real ( kind = 8 ), dimension (:, :), intent(in) coords,
    real ( kind = 8 ), dimension (:, :), intent(out) new_coords )

```

Apply translation and rotation to a set of atomic coordinates.

Transforms the coordinates of a complex from a frame centered at *xc2* to a new frame centered at *xc1*. The routine first recenters coordinates by *xc2*, applies a rotation defined by *rot_vx* and *rot_vy* (their cross product defines the third axis), then applies an additional translation *trans*, and writes the results to *new_coords*.

Parameters

in	<i>xc1</i>	Target center position (3)
in	<i>xc2</i>	Original center position (3)
in	<i>trans</i>	Translation to apply after rotation (3)
in	<i>rot_vx</i>	First rotation axis vector (3)
in	<i>rot_vy</i>	Second rotation axis vector (3)
in	<i>nb_atoms</i>	Number of atoms (integer)
in	<i>coords</i>	Input coordinates, shape (nb_atoms,3)
out	<i>new_coords</i>	Output coordinates, shape (nb_atoms,3)

10.1.2.6 vectors_angle_2d()

```

subroutine maths::vectors_angle_2d (
    real( kind=8 ), dimension( 3 ), intent(in) vector1,
    real( kind=8 ), dimension( 3 ), intent(in) vector2,
    real (kind=8), intent(out) theta_degrees )

```

Compute the angle in degrees between two vectors projected to 2D.

Uses only the first two components of the input vectors to compute the angle in the XY plane. Returns 0 if either vector has zero length in the XY plane.

Parameters

in	<i>vector1</i>	First vector (use components 1 and 2)
in	<i>vector2</i>	Second vector (use components 1 and 2)
out	<i>theta_degrees</i>	Angle between projected vectors in degrees

10.1.2.7 vectors_angle_3d()

```

subroutine maths::vectors_angle_3d (

```

```

real( kind=8 ), dimension( 3 ), intent(in) vector1,
real( kind=8 ), dimension( 3 ), intent(in) vector2,
real( kind=8), intent(out) theta_degrees )

```

Compute the angle in degrees between two 3D vectors.

Calculates the angle (in degrees) between `vector1` and `vector2` using the dot product formula and `acos`. No modification of inputs.

Parameters

in	<code>vector1</code>	First 3D vector (3 elements)
in	<code>vector2</code>	Second 3D vector (3 elements)
out	<code>theta_degrees</code>	Angle between vectors in degrees

10.2 mod_assoc Module Reference

Association/complexes file utilities.

Data Types

- type [type_assoc_file](#)
Data type for storing binned data (eg.

Functions/Subroutines

- subroutine [allocate_assoc_object](#) (this, nlines)
Allocate storage for an association/complexes object.
- integer function [size_assoc](#) (assoc_file)
Count encounter records in an association/complexes file.
- subroutine [fill_assoc_object](#) (this, assoc_file)
Read an association/complexes file and populate [type_assoc_file](#).

10.2.1 Detailed Description

Association/complexes file utilities.

This module defines the [type_assoc_file](#) derived type and routines to allocate and populate it from an association/complexes text file. The [type_assoc_file](#) stores the raw file lines, header lines, per-encounter translation vectors and rotation vectors, and the two reference centers `xc1` and `xc2` extracted from the file header.

NOTE: The routines in this file are adapted from the SDA7 project's [mod_pdb.f90](#) (original author Neil Bruce). This file contains code and ideas derived from that implementation; appropriate attribution to the SDA7 sources is retained here. Consult the SDA7 project for original authorship and licensing details.

Author

Abraham Muñiz-Chicharro

10.2.2 Function/Subroutine Documentation

10.2.2.1 allocate_assoc_object()

```

subroutine mod_assoc::allocate_assoc_object (
    type(type\_assoc\_file), intent(inout) this,
    integer, intent(in) nlines )

```

Allocate storage for an association/complexes object.

Allocates the arrays inside `this` to hold `nlines` encounter records and initializes numeric arrays to zero.

Parameters

in, out	<i>this</i>	Instance of type_assoc_file to initialize
in	<i>nlines</i>	Number of encounter lines to allocate

10.2.2.2 fill_assoc_object()

```
subroutine mod_assoc::fill_assoc_object (
    type(type\_assoc\_file), intent(inout) this,
    integer, intent(in) assoc_file )
```

Read an association/complexes file and populate [type_assoc_file](#).

Reads header lines from the open unit *assoc_file* and extracts the two reference centers (*xc1*, *xc2*) from header lines 3 and 4. Subsequent non-header lines are stored into *thislines* and the translation and rotation vectors are parsed into *thistrans_vector*, *thisrot1* and *thisrot2*.

Parameters

in, out	<i>this</i>	Allocated type_assoc_file to populate
in	<i>assoc_file</i>	Fortran unit number of an already-open file

10.2.2.3 size_assoc()

```
integer function mod_assoc::size_assoc (
    integer assoc_file )
```

Count encounter records in an association/complexes file.

Scans *assoc_file* (a Fortran unit number) and counts the number of non-comment lines. Lines starting with # are treated as comments and ignored. Returns the number of encounter records.

Parameters

in	<i>assoc_file</i>	Fortran unit number of an already-open file
----	-------------------	---

Returns

nlines Number of encounter records found

10.3 mod_clust_algorithm Module Reference**Functions/Subroutines**

- subroutine [linkage_clustering](#) (*matrix*, *n*, *linkage_type*, *output_name*, *complexes*, *opt_array*)
Perform hierarchical clustering on a distance matrix.
- subroutine [sort_complexes](#) (*tot_encounters*, *cluster_indexes*, *cluster_count*, *active_clusters*, *representative_indexes*, *cluster_average*, *cluster_sd*, *array*)
Sort clusters and associated arrays by a given key array.
- subroutine [write_cluster_elements](#) (*tot_encounters*, *cluster_indexes*, *cluster_count*, *active_clusters*, *output_name*)
Write cluster membership lists to a text file.
- subroutine [write_clust_info](#) (*tot_encounters*, *representative_indexes*, *active_clusters*, *clust_count*, *cluster_average*, *cluster_sd*, *output_name*)
Write summary information about clusters to a text file.

- subroutine `write_complexes` (tot_encounters, cluster_indexes, cluster_count, active_clusters, output_name, complexes)

Write complexes contained in each cluster to individual files.

10.3.1 Function/Subroutine Documentation

10.3.1.1 linkage_clustering()

```
subroutine mod_clust_algorithm::linkage_clustering (
    real (kind=8), dimension(:, :), intent(inout) matrix,
    integer, intent(in) n,
    character(len=*), intent(in) linkage_type,
    character*128, intent(in) output_name,
    type(type_assoc_file) complexes,
    real (kind=8), dimension(:), intent(inout), optional opt_array )
```

Perform hierarchical clustering on a distance matrix.

This routine performs hierarchical clustering using minimum, maximum, or mean linkage on the input `matrix` (shape `(n,n)`) of pairwise distances or similarity scores. The routine mutates `matrix` during merging and terminates merging when clusters are farther apart than an internally computed threshold. Results are written to files named using `output_name` and complexes information is optionally recorded via the `complexes` structure.

Parameters

in, out	<code>matrix</code>	Pairwise distance/similarity matrix (n,n). This matrix is updated during clustering.
in	<code>n</code>	Number of elements / encounters (integer).
in	<code>linkage_type</code>	Type of linkage: 'min', 'max', or 'mean' (string).
in	<code>output_name</code>	Base name for output files (string).
in	<code>complexes</code>	Structure containing complex text lines/headers.
in	<code>opt_array</code>	Optional array of per-encounter values used to compute cluster averages instead of reading values from <code>matrix</code> .

10.3.1.2 sort_complexes()

```
subroutine mod_clust_algorithm::sort_complexes (
    integer, intent(in) tot_encounters,
    integer, dimension(:, :), intent(inout) cluster_indexes,
    integer, dimension(:), intent(inout) cluster_count,
    logical, dimension(:), intent(inout) active_clusters,
    integer, dimension(:), intent(inout) representative_indexes,
    real (kind=8), dimension(:), intent(inout) cluster_average,
    real (kind=8), dimension(:), intent(inout) cluster_sd,
    real (kind=8), dimension(:), intent(inout) array )
```

Sort clusters and associated arrays by a given key array.

Reorders `cluster_indexes` and all supplied per-cluster arrays so that they are sorted by ascending values in `array`. This is a stable reordering helper used prior to writing results.

Parameters

in	<code>tot_encounters</code>	Number of clusters/encounters
in, out	<code>cluster_indexes</code>	Matrix storing cluster member indexes
in, out	<code>cluster_count</code>	Number of members in each cluster
in, out	<code>active_clusters</code>	Logical flags marking active clusters
in, out	<code>representative_indexes</code>	Representative index per cluster

Parameters

in, out	<i>cluster_average</i>	Per-cluster average values
in, out	<i>cluster_sd</i>	Per-cluster standard deviation
in, out	<i>array</i>	Key array used to sort clusters

10.3.1.3 write_clust_info()

```
subroutine mod_clust_algorithm::write_clust_info (
    integer, intent(in) tot_encounters,
    integer, dimension(:), intent(in) representative_indexes,
    logical, dimension(:), intent(in) active_clusters,
    integer, dimension(tot_encounters), intent(in) clust_count,
    real (kind=8), dimension(tot_encounters) cluster_average,
    real (kind=8), dimension(tot_encounters) cluster_sd,
    character*128, intent(in) output_name )
```

Write summary information about clusters to a text file.

Produces a summary file <output_name>_clust_info.txt listing cluster population, representative index, average and standard deviation for each active cluster.

Parameters

in	<i>tot_encounters</i>	Total number of encounters
in	<i>clust_count</i>	Array with population of each cluster
in	<i>active_clusters</i>	Logical mask indicating active clusters
in	<i>representative_indexes</i>	Representative index per cluster
in	<i>output_name</i>	Base filename for outputs
in	<i>cluster_average</i>	Per-cluster average values
in	<i>cluster_sd</i>	Per-cluster standard deviations

10.3.1.4 write_cluster_elements()

```
subroutine mod_clust_algorithm::write_cluster_elements (
    integer, intent(in) tot_encounters,
    integer, dimension(:, :), intent(in) cluster_indexes,
    integer, dimension(:), intent(in) cluster_count,
    logical, dimension(:), intent(in) active_clusters,
    character*128, intent(in) output_name )
```

Write cluster membership lists to a text file.

Outputs a file named <output_name>_clusters.txt containing the members of each active cluster in a matrix-like format.

Parameters

in	<i>tot_encounters</i>	Number of possible clusters/encounters
in	<i>cluster_indexes</i>	Matrix of cluster member indexes
in	<i>cluster_count</i>	Number of members per cluster
in	<i>active_clusters</i>	Logical mask indicating active clusters
in	<i>output_name</i>	Base filename for output

10.3.1.5 write_complexes()

```
subroutine mod_clust_algorithm::write_complexes (
    integer, intent(in) tot_encounters,
    integer, dimension(:, :), intent(in) cluster_indexes,
    integer, dimension(:), intent(in) cluster_count,
    logical, dimension(:), intent(in) active_clusters,
    character*128, intent(in) output_name,
    type(type_assoc_file) complexes )
```

Write complexes contained in each cluster to individual files.

For each active cluster this routine creates a file <output_name>_clusterXXXX_complexes containing the header lines from complexeshead and the encounter lines for the cluster members taken from complexeslines.

Parameters

in	<i>tot_encounters</i>	Number of clusters/encounters
in	<i>cluster_indexes</i>	Matrix of cluster member indexes
in	<i>cluster_count</i>	Number of members per cluster
in	<i>active_clusters</i>	Logical mask indicating active clusters
in	<i>output_name</i>	Base filename for output files
in	<i>complexes</i>	Structure with header and lines to write

10.4 mod_matrix Module Reference

Functions/Subroutines

- subroutine [matrix_z_coord](#) (matrix, array, n, nb_atoms, xc1, xc2, trans_vector, rot1, rot2, solute_crds)
Build a matrix of absolute differences of Z-coordinates per encounter.
- subroutine [matrix_atoms_dist](#) (matrix, array, n, nb_atoms, xc1, xc2, trans_vector, rot1, rot2, solute1_crds, solute2_crds)
Build matrix from minimum inter-atomic distances between two solutes.
- subroutine [matrix_angle](#) (matrix, array, n, nb_atoms, xc1, xc2, trans_vector, rot1, rot2, point1a, point1b, point2a, point2b, dimensions)
Build matrix of angular differences between encounters.
- subroutine [matrix_rmsd](#) (matrix, n, nb_atoms, xc1, xc2, trans_vector, rot1, rot2, coord)
Build RMSD-based pairwise matrix between transformed coordinates.
- subroutine [write_matrix](#) (matrix, filename)
Write a numeric matrix to a formatted text file.
- subroutine [write_array](#) (array, filename)
Write a numeric array to a formatted text file.
- subroutine [read_matrix](#) (matrix, n, filename)
Read a square numeric matrix from a formatted text file.
- subroutine [read_array](#) (array, n, filename)
Read a numeric array from a formatted text file.

10.4.1 Function/Subroutine Documentation

10.4.1.1 matrix_angle()

```

subroutine mod_matrix::matrix_angle (
    real(kind=8), dimension(n, n), intent(out) matrix,
    real(kind=8), dimension(n), intent(out) array,
    integer, intent(in) n,
    integer, intent(in) nb_atoms,
    real(kind=8), dimension(3), intent(in) xc1,
    real(kind=8), dimension(3), intent(in) xc2,
    real(kind=8), dimension(:, :), intent(in) trans_vector,
    real(kind=8), dimension(:, :), intent(in) rot1,
    real(kind=8), dimension(:, :), intent(in) rot2,
    real(kind=8), dimension(:), intent(in) point1a,
    real(kind=8), dimension(:), intent(in) point1b,
    real(kind=8), dimension(:), intent(in) point2a,
    real(kind=8), dimension(:), intent(in) point2b,
    integer, intent(in) dimensions )

```

Build matrix of angular differences between encounters.

Computes the angle (2D or 3D) for each encounter and fills the pairwise absolute difference matrix.

Parameters

out	<i>matrix</i>	Output (n,n) matrix of pairwise angular differences
out	<i>array</i>	Output (n) array with per-encounter angles
in	<i>n</i>	Number of encounters
in	<i>nb_atoms</i>	Number of atoms in solute 2 to transform
in	<i>xc1,xc2</i>	Solute centers used for translation transformations
in	<i>trans_vector</i>	Transform translation vectors (n,3)
in	<i>rot1,rot2</i>	Transform rotation arrays (n,3)
in	<i>point1a</i>	First point defining reference vector
in	<i>point1b</i>	Second point defining reference vector
in	<i>point2a</i>	First point of vector to be transformed
in	<i>point2b</i>	Second point of vector to be transformed
in	<i>dimensions</i>	Dimensionality of angle calculation (2 or 3)

10.4.1.2 matrix_atoms_dist()

```

subroutine mod_matrix::matrix_atoms_dist (
    real(kind=8), dimension(n, n), intent(out) matrix,
    real(kind=8), dimension(n), intent(out) array,
    integer, intent(in) n,
    integer, intent(in) nb_atoms,
    real(kind=8), dimension(3), intent(in) xc1,
    real(kind=8), dimension(3), intent(in) xc2,
    real(kind=8), dimension(:, :), intent(in) trans_vector,
    real(kind=8), dimension(:, :), intent(in) rot1,
    real(kind=8), dimension(:, :), intent(in) rot2,
    real(kind=8), dimension(:, :), intent(in) solute1_crds,
    real(kind=8), dimension(:, :), intent(in) solute2_crds )

```

Build matrix from minimum inter-atomic distances between two solutes.

For each encounter, the second solute is transformed and the minimum distance to *solute1_crds* is computed; pairwise absolute differences of these minima populate *matrix*.

Parameters

out	<i>matrix</i>	Output (n,n) matrix of pairwise values
out	<i>array</i>	Output (n) array with per-encounter minimum distances
in	<i>n</i>	Number of encounters
in	<i>nb_atoms</i>	Number of atoms in solute 2 to transform
in	<i>xc1,xc2</i>	Solute centers used for translation transformations
in	<i>trans_vector</i>	Transform translation vectors (n,3)
in	<i>rot1,rot2</i>	Transform rotation arrays (n,3)
in	<i>solute1_crds</i>	Coordinates of first solute atoms
in	<i>solute2_crds</i>	Coordinates of second solute atoms

10.4.1.3 **matrix_rmsd()**

```

subroutine mod_matrix::matrix_rmsd (
    real(kind=8), dimension(n, n), intent(out) matrix,
    integer, intent(in) n,
    integer, intent(in) nb_atoms,
    real(kind=8), dimension(3), intent(in) xc1,
    real(kind=8), dimension(3), intent(in) xc2,
    real(kind=8), dimension(:, :), intent(in) trans_vector,
    real(kind=8), dimension(:, :), intent(in) rot1,
    real(kind=8), dimension(:, :), intent(in) rot2,
    real(kind=8), dimension(:, :), intent(in) coord )

```

Build RMSD-based pairwise matrix between transformed coordinates.

Applies transformation to the coordinate set for each encounter and computes pairwise RMSD values between transformed coordinate sets.

Parameters

out	<i>matrix</i>	Output (n,n) matrix of pairwise RMSD values
in	<i>n</i>	Number of encounters
in	<i>nb_atoms</i>	Number of atoms in solute 2 to transform
in	<i>xc1,xc2</i>	Solute centers used for translation transformations
in	<i>trans_vector</i>	Transform translation vectors (n,3)
in	<i>rot1,rot2</i>	Transform rotation arrays (n,3)
in	<i>coord</i>	Coordinates to be transformed

10.4.1.4 **matrix_z_coord()**

```

subroutine mod_matrix::matrix_z_coord (
    real(kind=8), dimension(n, n), intent(out) matrix,
    real(kind=8), dimension(n), intent(out) array,
    integer, intent(in) n,
    integer, intent(in) nb_atoms,
    real(kind=8), dimension(3), intent(in) xc1,
    real(kind=8), dimension(3), intent(in) xc2,
    real(kind=8), dimension(:, :), intent(in) trans_vector,
    real(kind=8), dimension(:, :), intent(in) rot1,
    real(kind=8), dimension(:, :), intent(in) rot2,
    real(kind=8), dimension(:, :), intent(in) solute_crds )

```

Build a matrix of absolute differences of Z-coordinates per encounter.

Computes the Z coordinate for each encounter (as in `array_z_coord`) and fills `matrix(i,j) = |z_i - z_j|` for $i < j$ (symmetrically).

Parameters

out	<i>matrix</i>	Output (n,n) matrix of pairwise values
out	<i>array</i>	Output array with per-encounter Z values
in	<i>n</i>	Number of encounters
in	<i>nb_atoms</i>	Number of atoms in <code>solute_crds</code>
in	<i>xc1,xc2</i>	Solute centers used for translation transformations
in	<i>trans_vector,rot1,rot2</i>	Transform arrays (n,3)
in	<i>solute_crds</i>	Coordinates of solute atoms

10.4.1.5 read_array()

```
subroutine mod_matrix::read_array (
    real(kind=8), dimension(:), intent(out), allocatable array,
    integer, intent(inout) n,
    character*128, intent(in) filename )
```

Read a numeric array from a formatted text file.

Parses a single-row formatted array file produced by `write_array`.

Parameters

out	<i>array</i>	Output array read from file (allocated)
in, out	<i>n</i>	Number of encounters (adjusted if file is smaller)
in	<i>filename</i>	Input filename

10.4.1.6 read_matrix()

```
subroutine mod_matrix::read_matrix (
    real(kind=8), dimension(:, :), intent(out), allocatable matrix,
    integer, intent(inout) n,
    character*128, intent(in) filename )
```

Read a square numeric matrix from a formatted text file.

Reads the file line-by-line to determine matrix size and parses rows with the expected F10.4 formatting.

Parameters

out	<i>matrix</i>	Output matrix read from file (allocated)
in, out	<i>n</i>	Number of encounters (adjusted if file is smaller)
in	<i>filename</i>	Input filename

10.4.1.7 write_array()

```
subroutine mod_matrix::write_array (
    real(kind=8), dimension(:), intent(in) array,
    character*128, intent(in) filename )
```

Write a numeric array to a formatted text file.

Writes `array` as a single formatted line using F10.4 floats.

Parameters

in	<i>array</i>	Input array to write
in	<i>filename</i>	Output filename

10.4.1.8 write_matrix()

```
subroutine mod_matrix::write_matrix (
    real(kind=8), dimension(:, :), intent(in) matrix,
    character*128, intent(in) filename )
```

Write a numeric matrix to a formatted text file.

Writes *matrix* with each row as F10.4 formatted floats. The output filename is provided by *filename*.

Parameters

in	<i>matrix</i>	Input matrix to write
in	<i>filename</i>	Output filename

10.5 mod_pdb Module Reference

Module to define a pdb type and pdb reading routine for tools.

Data Types

- type [type_pdb_atom](#)
- type [type_pdb_chain](#)
- type [type_pdb_file](#)
Data type for storing binned data (eg.
- type [type_pdb_residue](#)

Functions/Subroutines

- subroutine [allocate_pdb_object](#) (this, pdb_file, natoms, nresidues, nchains)
Allocate and initialize a [type_pdb_file](#) structure from a PDB file.
- integer function [count_atoms](#) (pdb_file)
Count ATOM/HETATM records in a PDB file.
- integer function [count_residues](#) (pdb_file)
Count residues in a PDB file.
- integer function [count_chains](#) (pdb_file)
Count chain separators (TER records) in a PDB file.
- subroutine [fill_pdb_object](#) (this, pdb_file)
Parse PDB records and populate the [type_pdb_file](#) structure.

10.5.1 Detailed Description

Module to define a pdb type and pdb reading routine for tools.

NOTE: This implementation is based on the SDA7 project's [mod_pdb.f90](#) (original author Neil Bruce). The present file has been modified and expanded for the clustering tools in this workspace. Notable enhancements include the addition of `count_residues` and `count_chains` functions and changes that allow creation of richer in-memory objects where residues contain their atoms and chains contain residues and atoms. These changes make it possible to build [type_pdb_file](#) instances with proper `residues` and `chains` substructures (each storing their atoms) rather than storing atoms in a single flat list only.

The original SDA7 code and author attribution are retained; consult the SDA7 sources for original licensing and authorship details.

Author

Neil Bruce @modified-by Abraham Muñiz-Chicharro

10.5.2 Function/Subroutine Documentation

10.5.2.1 allocate_pdb_object()

```
subroutine mod_pdb::allocate_pdb_object (
    type(type_pdb_file), intent(inout) this,
    integer, intent(in) pdb_file,
    integer, intent(in) natoms,
    integer, intent(in) nresidues,
    integer, intent(in) nchains )
```

Allocate and initialize a `type_pdb_file` structure from a PDB file.

This routine allocates the storage for `thisatoms`, `thisresidues` and `thischains` using the supplied counts and performs a preliminary scan of the open Fortran unit `pdb_file` to determine per-chain and per-residue allocation sizes for nested arrays.

Parameters

in, out	<i>this</i>	<code>type_pdb_file</code> instance to allocate and initialize
in	<i>pdb_file</i>	Fortran unit number of an already-open PDB file
in	<i>natoms</i>	Total number of ATOM/HETATM records expected
in	<i>nresidues</i>	Number of residues expected
in	<i>nchains</i>	Number of chains expected

10.5.2.2 count_atoms()

```
integer function mod_pdb::count_atoms (
    integer pdb_file )
```

Count ATOM/HETATM records in a PDB file.

Scans the open Fortran file unit `pdb_file` and returns the number of lines starting with ATOM or HETATM.

Parameters

in	<i>pdb_file</i>	Fortran unit number of an already-open PDB file
----	-----------------	---

Returns

`natoms` Number of ATOM/HETATM records found

10.5.2.3 count_chains()

```
integer function mod_pdb::count_chains (
    integer pdb_file )
```

Count chain separators (TER records) in a PDB file.

Scans the open Fortran file unit `pdb_file` and counts the number of TER records, which are treated as chain separators. Returns the number of chains found.

Parameters

in	<i>pdb_file</i>	Fortran unit number of an already-open PDB file
----	-----------------	---

Returns

nchains Number of chains (TER records) found

10.5.2.4 count_residues()

```
integer function mod_pdb::count_residues (
    integer pdb_file )
```

Count residues in a PDB file.

Scans the open Fortran file unit *pdb_file* and counts residue changes by comparing residue identifiers in ATOM/HETATM records. Returns the number of distinct residues found.

Parameters

in	<i>pdb_file</i>	Fortran unit number of an already-open PDB file
----	-----------------	---

Returns

nresidues Number of residues found

10.5.2.5 fill_pdb_object()

```
subroutine mod_pdb::fill_pdb_object (
    type(type_pdb_file), intent(inout) this,
    integer, intent(in) pdb_file )
```

Parse PDB records and populate the *type_pdb_file* structure.

Reads ATOM/HETATM records from the already-open Fortran unit *pdb_file*, fills *this* atoms with atom records and populates *this*residues and *this*chains so that residues contain their per-residue atoms arrays and chains contain their residues and atoms. The routine expects *this* to have been allocated with the correct sizes (see *allocate_pdb_object*).

Parameters

in, out	<i>this</i>	Pre-allocated <i>type_pdb_file</i> to populate
in	<i>pdb_file</i>	Fortran unit number of an already-open PDB file

10.6 mod_threshold Module Reference

Functions/Subroutines

- subroutine [array_z_coord](#) (array, n, nb_atoms, xc1, xc2, trans_vector, rot1, rot2, solute_crds)
Compute the Z-coordinate (along transformed z-axis) for each encounter.
- subroutine [array_atoms_dist](#) (array, n, nb_atoms, xc1, xc2, trans_vector, rot1, rot2, solute1_crds, solute2_crds)
Compute minimum inter-atomic distance between two solutes for each encounter.
- subroutine [array_angle](#) (array, n, nb_atoms, xc1, xc2, trans_vector, rot1, rot2, point1a, point1b, point2a, point2b, dimensions)
Compute angle between two vectors (defined by atom groups) per encounter.

- subroutine `sort_array` (arr, encounter_indexes)

Sort an array in ascending order and reorder encounter indexes accordingly.

10.6.1 Function/Subroutine Documentation

10.6.1.1 array_angle()

```
subroutine mod_threshold::array_angle (
    real(kind=8), dimension(n), intent(out) array,
    integer, intent(in) n,
    integer, intent(in) nb_atoms,
    real(kind=8), dimension(3), intent(in) xc1,
    real(kind=8), dimension(3), intent(in) xc2,
    real(kind=8), dimension(:, :), intent(in) trans_vector,
    real(kind=8), dimension(:, :), intent(in) rot1,
    real(kind=8), dimension(:, :), intent(in) rot2,
    real(kind=8), dimension(:), intent(in) point1a,
    real(kind=8), dimension(:), intent(in) point1b,
    real(kind=8), dimension(:), intent(in) point2a,
    real(kind=8), dimension(:), intent(in) point2b,
    integer, intent(in) dimensions )
```

Compute angle between two vectors (defined by atom groups) per encounter.

Transforms the pair of points from solute2 for each encounter and computes the angle between the vector defined by point1b-point1a and the vector defined by the transformed point2b-point2a.

Parameters

out	<i>array</i>	Output array with angles in degrees (n)
in	<i>n</i>	Number of encounters
in	<i>nb_atoms</i>	Number of atoms used for transformation
in	<i>xc1,xc2</i>	Centers used for transform
in	<i>trans_vector,rot1,rot2</i>	Transform arrays (n,3)
in	<i>point1a,b,point2a,b</i>	Reference points/vectors
in	<i>dimensions</i>	Use 2 or 3 to select projection

10.6.1.2 array_atoms_dist()

```
subroutine mod_threshold::array_atoms_dist (
    real(kind=8), dimension(n), intent(out) array,
    integer, intent(in) n,
    integer, intent(in) nb_atoms,
    real(kind=8), dimension(3), intent(in) xc1,
    real(kind=8), dimension(3), intent(in) xc2,
    real(kind=8), dimension(:, :), intent(in) trans_vector,
    real(kind=8), dimension(:, :), intent(in) rot1,
    real(kind=8), dimension(:, :), intent(in) rot2,
    real(kind=8), dimension(:, :), intent(in) solute1_crds,
    real(kind=8), dimension(:, :), intent(in) solute2_crds )
```

Compute minimum inter-atomic distance between two solutes for each encounter.

For each encounter the second solute is transformed and the minimum distance between any atom of solute1←_crds and the transformed solute2_crds is computed and stored in array.

Parameters

out	<i>array</i>	Output array (n)
in	<i>n</i>	Number of encounters
in	<i>nb_atoms</i>	Number of atoms in <i>solute2_crds</i>
in	<i>xc1,xc2</i>	Centers used for transform
in	<i>trans_vector,rot1,rot2</i>	Transform arrays (n,3)
in	<i>solute1_crds</i>	Coordinates of solute1 atoms
in	<i>solute2_crds</i>	Coordinates of solute2 atoms

10.6.1.3 `array_z_coord()`

```

subroutine mod_threshold::array_z_coord (
    real(kind=8), dimension(n), intent(out) array,
    integer, intent(in) n,
    integer, intent(in) nb_atoms,
    real(kind=8), dimension(3), intent(in) xc1,
    real(kind=8), dimension(3), intent(in) xc2,
    real(kind=8), dimension(:, :), intent(in) trans_vector,
    real(kind=8), dimension(:, :), intent(in) rot1,
    real(kind=8), dimension(:, :), intent(in) rot2,
    real(kind=8), dimension(:, :), intent(in) solute_crds )

```

Compute the Z-coordinate (along transformed z-axis) for each encounter.

Recenters each encounter at *xc2*, applies rotation/translation using *rot1/rot2* and *trans_vector*, and extracts the Z coordinate of the first atom in *solute_crds* for every encounter into *array*.

Parameters

out	<i>array</i>	Output array of length n with z-coordinates
in	<i>n</i>	Number of encounters
in	<i>nb_atoms</i>	Number of atoms in <i>solute_crds</i>
in	<i>xc1,xc2</i>	Centers used for transform
in	<i>trans_vector,rot1,rot2</i>	Transform arrays (n,3)
in	<i>solute_crds</i>	Coordinates of solute atoms (nb_atoms,3)

10.6.1.4 `sort_array()`

```

subroutine mod_threshold::sort_array (
    real (kind=8), dimension(:), intent(inout) arr,
    integer, dimension(:), intent(inout) encounter_indexes )

```

Sort an array in ascending order and reorder encounter indexes accordingly.

Simple selection sort used to reorder *arr* and the parallel *encounter_indexes* array. Intended for small to moderate sizes.

Parameters

in, out	<i>arr</i>	Array to sort (modified in-place)
in, out	<i>encounter_indexes</i>	Parallel index array reordered to match <i>arr</i>

10.7 read_input Module Reference

Functions/Subroutines

- subroutine `read_pdb` (pdb, filename, tot_atoms, tot_residues, tot_chains)
Read a PDB file into a `type_pdb_file` object.
- subroutine `read_assoc` (assoc, filename, tot_encounters)
Read an association/complexes file into a `type_assoc_file` object.

10.7.1 Function/Subroutine Documentation

10.7.1.1 read_assoc()

```
subroutine read_input::read_assoc (
    type ( type_assoc_file ) assoc,
    character*128, intent(in) filename,
    integer, intent(out) tot_encounters )
```

Read an association/complexes file into a `type_assoc_file` object.

Checks the file exists, opens it and uses `size_assoc`, `allocate_assoc_object` and `fill_assoc_object` to construct the `assoc` structure.

Parameters

out	<i>assoc</i>	Allocated and filled assoc object
in	<i>filename</i>	Path to complexes file
out	<i>tot_encounters</i>	Number of encounter records read

10.7.1.2 read_pdb()

```
subroutine read_input::read_pdb (
    type ( type_pdb_file ) pdb,
    character*128, intent(in) filename,
    integer, intent(out) tot_atoms,
    integer, intent(out) tot_residues,
    integer, intent(out) tot_chains )
```

Read a PDB file into a `type_pdb_file` object.

Validates the existence of `filename`, opens it, determines the counts for atoms, residues and chains, allocates `pdb` and fills it using `allocate_pdb_object` and `fill_pdb_object`.

Parameters

out	<i>pdb</i>	Allocated and filled PDB object
in	<i>filename</i>	Path to PDB file
out	<i>tot_atoms</i>	Number of atoms found
out	<i>tot_residues</i>	Number of residues found
out	<i>tot_chains</i>	Number of chains found

Chapter 11

Data Type Documentation

11.1 `mod_assoc::type_assoc_file` Type Reference

Data type for storing binned data (eg.

Public Attributes

- integer `nlines`
Number of lines in assoc_file file.
- character(len=217), dimension(:), allocatable `lines`
- character(len=217), dimension(4) `head`
- real(kind=8), dimension(:, :), allocatable `trans_vector`
Translational vector.
- real(kind=8), dimension(:, :), allocatable `rot1`
rotX, rotY
- real(kind=8), dimension(:, :), allocatable `rot2`
- real(kind=8), dimension(3) `xc1`
center of mass of p2
- real(kind=8), dimension(3) `xc2`

11.1.1 Detailed Description

Data type for storing binned data (eg.
Radial distribution functions)

11.1.2 Member Data Documentation

11.1.2.1 `head`

```
character(len=217), dimension(4) mod_assoc::type_assoc_file::head
```

11.1.2.2 `lines`

```
character(len=217), dimension(:), allocatable mod_assoc::type_assoc_file::lines
```

11.1.2.3 `nlines`

```
integer mod_assoc::type_assoc_file::nlines
```

Number of lines in assoc_file file.

11.1.2.4 rot1

```
real (kind=8), dimension (:,:), allocatable mod_assoc::type_assoc_file::rot1
rotX, rotY
```

11.1.2.5 rot2

```
real (kind=8), dimension (:,:), allocatable mod_assoc::type_assoc_file::rot2
```

11.1.2.6 trans_vector

```
real (kind=8), dimension (:, :), allocatable mod_assoc::type_assoc_file::trans_vector
Translational vector.
```

11.1.2.7 xc1

```
real (kind=8), dimension (3) mod_assoc::type_assoc_file::xc1
center of mass of p2
```

11.1.2.8 xc2

```
real (kind=8), dimension (3) mod_assoc::type_assoc_file::xc2
The documentation for this type was generated from the following file:
```

- [mod_assoc.f90](#)

11.2 mod_pdb::type_pdb_atom Type Reference

Public Attributes

- integer [record](#)
Number of atoms in pdb file Flag for record type, 1 = ATOM, 2 = HETATM.
- real(kind=8), dimension(3) [coord](#)
Position, occupancy and temp factor - can be used to store other data eg in pqr.
- real(kind=8) [occ](#)
- real(kind=8) [beta](#)
- character(len=4) [name](#)
Atom name, res name and chain name.
- character(len=3) [resname](#)
- character(len=1) [chainid](#)
- integer [id](#)
Atom and residue numbers.
- integer [resid](#)

11.2.1 Member Data Documentation

11.2.1.1 beta

```
real (kind=8) mod_pdb::type_pdb_atom::beta
```

11.2.1.2 chainid

character (len=1) mod_pdb::type_pdb_atom::chainid

11.2.1.3 coord

real (kind=8), dimension (3) mod_pdb::type_pdb_atom::coord

Position, occupancy and temp factor - can be used to store other data eg in pqr.

11.2.1.4 id

integer mod_pdb::type_pdb_atom::id

Atom and residue numbers.

11.2.1.5 name

character (len=4) mod_pdb::type_pdb_atom::name

Atom name, res name and chain name.

11.2.1.6 occ

real (kind=8) mod_pdb::type_pdb_atom::occ

11.2.1.7 record

integer mod_pdb::type_pdb_atom::record

Number of atoms in pdb file Flag for record type, 1 = ATOM, 2 = HETATM.

11.2.1.8 resid

integer mod_pdb::type_pdb_atom::resid

11.2.1.9 resname

character (len=3) mod_pdb::type_pdb_atom::resname

The documentation for this type was generated from the following file:

- [mod_pdb.f90](#)

11.3 mod_pdb::type_pdb_chain Type Reference

Collaboration diagram for mod_pdb::type_pdb_chain:

11.4 mod_pdb::type_pdb_file Type Reference

Data type for storing binned data (eg.

Collaboration diagram for mod_pdb::type_pdb_file:

Public Attributes

- integer [natoms](#)
Number of atoms in pdb file.
- integer [nresidues](#)
- integer [nchains](#)
- type([type_pdb_atom](#)), dimension(:), allocatable [atoms](#)
Flag for record type, 1 = ATOM, 2 = HETATM.
- type([type_pdb_residue](#)), dimension(:), allocatable [residues](#)
- type([type_pdb_chain](#)), dimension(:), allocatable [chains](#)

11.4.1 Detailed Description

Data type for storing binned data (eg.
Radial distribution functions)

11.4.2 Member Data Documentation

11.4.2.1 atoms

`type(type\_pdb\_atom), dimension (:), allocatable mod_pdb::type_pdb_file::atoms`
Flag for record type, 1 = ATOM, 2 = HETATM.

11.4.2.2 chains

`type(type\_pdb\_chain), dimension (:), allocatable mod_pdb::type_pdb_file::chains`

11.4.2.3 natoms

`integer mod_pdb::type_pdb_file::natoms`
Number of atoms in pdb file.

11.4.2.4 nchains

`integer mod_pdb::type_pdb_file::nchains`

11.4.2.5 nresidues

`integer mod_pdb::type_pdb_file::nresidues`

11.4.2.6 residues

`type(type\_pdb\_residue), dimension (:), allocatable mod_pdb::type_pdb_file::residues`
The documentation for this type was generated from the following file:

- [mod_pdb.f90](#)

11.5 mod_pdb::type_pdb_residue Type Reference

Collaboration diagram for mod_pdb::type_pdb_residue:

Public Attributes

- integer [natoms](#)
Number of atoms in pdb file.
- character [chainid](#)
Chain id.
- character(len=3) [resname](#)
Residue name.
- integer [resid](#)
Residue number.
- type([type_pdb_atom](#)), dimension(:), allocatable [atoms](#)
Flag for record type, 1 = ATOM, 2 = HETATM.

11.5.1 Member Data Documentation

11.5.1.1 atoms

`type(type\_pdb\_atom), dimension (:), allocatable mod_pdb::type_pdb_residue::atoms`
Flag for record type, 1 = ATOM, 2 = HETATM.

11.5.1.2 chainid

`character mod_pdb::type_pdb_residue::chainid`
Chain id.

11.5.1.3 natoms

`integer mod_pdb::type_pdb_residue::natoms`
Number of atoms in pdb file.

11.5.1.4 resid

`integer mod_pdb::type_pdb_residue::resid`
Residue number.

11.5.1.5 resname

`character (len=3) mod_pdb::type_pdb_residue::resname`
Residue name.

The documentation for this type was generated from the following file:

- [mod_pdb.f90](#)

Chapter 12

File Documentation

12.1 analyze_residues.f90 File Reference

Functions/Subroutines

- program [main](#)
Analyze which residues are close to the surface across encounters.
- subroutine [print_help](#) ()
Print usage/help text for analyze_residues.

12.1.1 Function/Subroutine Documentation

12.1.1.1 main()

program main

Analyze which residues are close to the surface across encounters.

This program reads a PDB for solute2 and an associations/complexes file, transforms residue centers for each encounter and counts how many encounters bring residues within a distance threshold from the surface. Results are written to `closest_residues.txt`.

Author

Abraham Muñoz-Chicharro

12.1.1.2 print_help()

subroutine main::print_help

Print usage/help text for `analyze_residues`.

Shows required command-line options and examples for running the residue analysis tool.

12.2 clust.f90 File Reference

Functions/Subroutines

- program [main](#)
Command-line program to perform hierarchical clustering on encounter matrices.
- subroutine [print_help](#) ()
Print usage and help information for the `clust` program.

12.2.1 Function/Subroutine Documentation

12.2.1.1 main()

program main

Command-line program to perform hierarchical clustering on encounter matrices.

Reads a pairwise matrix (and optional array) and runs hierarchical clustering with user-specified linkage method (min, max, or mean). Results are written to files using the `-output_name` base name. See `print_help` below for usage.

Author

Abraham Muñoz-Chicharro

12.2.1.2 print_help()

subroutine main::print_help

Print usage and help information for the `clust` program.

Describes command-line options and provides an example invocation.

12.3 make_matrix.f90 File Reference

Functions/Subroutines

- program [main](#)
Command-line utility to build encounter-based matrices.
- subroutine [print_help](#) (`help_option`)
Print usage/help text for `make_matrix`.
- subroutine [read_atoms_coord](#) (`atoms_array`, `atoms_coord`, `tot_atoms`, `pdb`, `pdb_filename`)
Read coordinates for a specified list of atoms (names or indices).

12.3.1 Function/Subroutine Documentation

12.3.1.1 main()

program main

Command-line utility to build encounter-based matrices.

Usage: see `print_help` below. This program reads PDB(s) and an association/complex file and generates matrices (`rmsd`, `z_coord`, `atoms_dist`, `angle`) and arrays derived from encounter transforms.

Author

Abraham Muñoz-Chicharro

12.3.1.2 print_help()

```
subroutine main::print_help (
    character(len=*), intent(in) help_option )
```

Print usage/help text for `make_matrix`.

Provides short usage examples for the main program and for each supported matrix type.

12.3.1.3 read_atoms_coord()

```
subroutine main::read_atoms_coord (
    character(len=*), dimension(:), intent(in) atoms_array,
    real(kind=8), dimension(:, :), intent(out), allocatable atoms_coord,
    integer, intent(in) tot_atoms,
    type ( type_pdb_file ), intent(in) pdb,
    character(len=*), intent(in) pdb_filename )
```

Read coordinates for a specified list of atoms (names or indices).

Accepts either atom indices or atom names. If atom names are provided, all matching atoms in the PDB are returned (in order).

Parameters

in	<i>atoms_array</i>	Array of atom name strings or index strings
in	<i>pdb_filename</i>	Reference filename for warnings
in	<i>tot_atoms</i>	Total number of atoms in pdb
out	<i>atoms_coord</i>	Allocated array with atom coordinates (N,3)

12.4 maths.f90 File Reference

Modules

- module [maths](#)
Mathematical utilities and coordinate transformation routines.

Functions/Subroutines

- subroutine [maths::cross](#) (v1, v2, v3)
Compute the cross product of two 3D vectors.
- subroutine [maths::update_complex](#) (xc1, xc2, trans, rot_vx, rot_vy, nb_atoms, coords, new_coords)
Apply translation and rotation to a set of atomic coordinates.
- subroutine [maths::vectors_angle_3d](#) (vector1, vector2, theta_degrees)
Compute the angle in degrees between two 3D vectors.
- subroutine [maths::vectors_angle_2d](#) (vector1, vector2, theta_degrees)
Compute the angle in degrees between two vectors projected to 2D.
- subroutine [maths::rmsd](#) (rmsd_value, nb_atoms, coords1, coords2)
Compute the root-mean-square deviation (RMSD) between two sets of coordinates.
- subroutine [maths::calculate_cog](#) (cog, coords, nb_atoms)
Compute the center of geometry (centroid) of a set of points.
- subroutine [maths::calculate_distance](#) (p1, p2, dist)
Compute Euclidean distance between two 3D points.

12.5 mod_assoc.f90 File Reference

Data Types

- type [mod_assoc::type_assoc_file](#)
Data type for storing binned data (eg.

Modules

- module [mod_assoc](#)
Association/complexes file utilities.

Functions/Subroutines

- subroutine [mod_assoc::allocate_assoc_object](#) (this, nlines)
Allocate storage for an association/complexes object.
- integer function [mod_assoc::size_assoc](#) (assoc_file)
Count encounter records in an association/complexes file.
- subroutine [mod_assoc::fill_assoc_object](#) (this, assoc_file)
Read an association/complexes file and populate [type_assoc_file](#).

12.6 mod_clust_algorithm.f90 File Reference

Modules

- module [mod_clust_algorithm](#)

Functions/Subroutines

- subroutine [mod_clust_algorithm::linkage_clustering](#) (matrix, n, linkage_type, output_name, complexes, opt←_array)
Perform hierarchical clustering on a distance matrix.
- subroutine [mod_clust_algorithm::sort_complexes](#) (tot_encounters, cluster_indexes, cluster_count, active_←clusters, representative_indexes, cluster_average, cluster_sd, array)
Sort clusters and associated arrays by a given key array.
- subroutine [mod_clust_algorithm::write_cluster_elements](#) (tot_encounters, cluster_indexes, cluster_count, active_clusters, output_name)
Write cluster membership lists to a text file.
- subroutine [mod_clust_algorithm::write_clust_info](#) (tot_encounters, representative_indexes, active_clusters, clust_count, cluster_average, cluster_sd, output_name)
Write summary information about clusters to a text file.
- subroutine [mod_clust_algorithm::write_complexes](#) (tot_encounters, cluster_indexes, cluster_count, active_←clusters, output_name, complexes)
Write complexes contained in each cluster to individual files.

12.7 mod_matrix.f90 File Reference

Modules

- module [mod_matrix](#)

Functions/Subroutines

- subroutine [mod_matrix::matrix_z_coord](#) (matrix, array, n, nb_atoms, xc1, xc2, trans_vector, rot1, rot2, solute_crds)
Build a matrix of absolute differences of Z-coordinates per encounter.
- subroutine [mod_matrix::matrix_atoms_dist](#) (matrix, array, n, nb_atoms, xc1, xc2, trans_vector, rot1, rot2, solute1_crds, solute2_crds)
Build matrix from minimum inter-atomic distances between two solutes.
- subroutine [mod_matrix::matrix_angle](#) (matrix, array, n, nb_atoms, xc1, xc2, trans_vector, rot1, rot2, point1a, point1b, point2a, point2b, dimensions)
Build matrix of angular differences between encounters.
- subroutine [mod_matrix::matrix_rmsd](#) (matrix, n, nb_atoms, xc1, xc2, trans_vector, rot1, rot2, coord)
Build RMSD-based pairwise matrix between transformed coordinates.
- subroutine [mod_matrix::write_matrix](#) (matrix, filename)
Write a numeric matrix to a formatted text file.

- subroutine `mod_matrix::write_array` (array, filename)
Write a numeric array to a formatted text file.
- subroutine `mod_matrix::read_matrix` (matrix, n, filename)
Read a square numeric matrix from a formatted text file.
- subroutine `mod_matrix::read_array` (array, n, filename)
Read a numeric array from a formatted text file.

12.8 mod_pdb.f90 File Reference

Data Types

- type `mod_pdb::type_pdb_file`
Data type for storing binned data (eg.
- type `mod_pdb::type_pdb_chain`
- type `mod_pdb::type_pdb_residue`
- type `mod_pdb::type_pdb_atom`

Modules

- module `mod_pdb`
Module to define a pdb type and pdb reading routine for tools.

Functions/Subroutines

- subroutine `mod_pdb::allocate_pdb_object` (this, pdb_file, natoms, nresidues, nchains)
Allocate and initialize a `type_pdb_file` structure from a PDB file.
- integer function `mod_pdb::count_atoms` (pdb_file)
Count ATOM/HETATM records in a PDB file.
- integer function `mod_pdb::count_residues` (pdb_file)
Count residues in a PDB file.
- integer function `mod_pdb::count_chains` (pdb_file)
Count chain separators (TER records) in a PDB file.
- subroutine `mod_pdb::fill_pdb_object` (this, pdb_file)
Parse PDB records and populate the `type_pdb_file` structure.

12.9 mod_threshold.f90 File Reference

Modules

- module `mod_threshold`

Functions/Subroutines

- subroutine `mod_threshold::array_z_coord` (array, n, nb_atoms, xc1, xc2, trans_vector, rot1, rot2, solute_crds)
Compute the Z-coordinate (along transformed z-axis) for each encounter.
- subroutine `mod_threshold::array_atoms_dist` (array, n, nb_atoms, xc1, xc2, trans_vector, rot1, rot2, solute1 ↔ _crds, solute2_crds)
Compute minimum inter-atomic distance between two solutes for each encounter.
- subroutine `mod_threshold::array_angle` (array, n, nb_atoms, xc1, xc2, trans_vector, rot1, rot2, point1a, point1b, point2a, point2b, dimensions)
Compute angle between two vectors (defined by atom groups) per encounter.
- subroutine `mod_threshold::sort_array` (arr, encounter_indexes)
Sort an array in ascending order and reorder encounter indexes accordingly.

12.10 read_input.f90 File Reference

Modules

- module `read_input`

Functions/Subroutines

- subroutine `read_input::read_pdb` (pdb, filename, tot_atoms, tot_residues, tot_chains)
Read a PDB file into a `type_pdb_file` object.
- subroutine `read_input::read_assoc` (assoc, filename, tot_encounters)
Read an association/complexes file into a `type_assoc_file` object.

12.11 threshold.f90 File Reference

Functions/Subroutines

- program `main`
Generate encounter selection based on threshold criteria.
- subroutine `print_help` (help_option)
Print usage/help text for `threshold`.
- subroutine `read_atoms_coord` (atoms_array, atoms_coord, tot_atoms, pdb, pdb_filename)
Read coordinates for a specified list of atoms (names or indices).
- subroutine `write_complexes` (complexes_arg, distarray_arg, encounter_indexes_arg, nb_encounters_arg, cutoff_arg, output_name_arg)
Write selected encounter complexes to an output file.
- subroutine `write_array` (array_arg, filename_arg, cutoff_arg)
Write array values up to a cutoff to a formatted file.

12.11.1 Function/Subroutine Documentation

12.11.1.1 main()

```
program main
```

Generate encounter selection based on threshold criteria.

This executable computes thresholds (z_coord, atoms_dist, angles) across encounter transforms and writes selected encounter records to an output complexes file. See `print_help` for usage examples.

Author

Abraham Muñoz-Chicharro

12.11.1.2 print_help()

```
subroutine main::print_help (
    character(len=*), intent(in) help_option )
```

Print usage/help text for `threshold`.

Explains command-line options and provides examples per threshold type.

12.11.1.3 read_atoms_coord()

```
subroutine main::read_atoms_coord (
    character(len=*), dimension(:), intent(in) atoms_array,
    real(kind=8), dimension(:, :), intent(out), allocatable atoms_coord,
    integer, intent(in) tot_atoms,
    type ( type_pdb_file ), intent(in) pdb,
    character(len=*), intent(in) pdb_filename )
```

Read coordinates for a specified list of atoms (names or indices).

Similar to the helper in `make_matrix`. Returns an allocated `atoms_coord` array with matching coordinates for the atom names or indices provided in `atoms_array`.

12.11.1.4 write_array()

```
subroutine main::write_array (
    real(kind=8), dimension(:), intent(in) array_arg,
    character*128, intent(in) filename_arg,
    real(kind=8), intent(in) cutoff_arg )
```

Write array values up to a cutoff to a formatted file.

Writes values from `array` (in ascending order) until `cutoff` is reached and stores them in `filename` using F10.4 formatting.

12.11.1.5 write_complexes()

```
subroutine main::write_complexes (
    type(type_assoc_file), intent(in) complexes_arg,
    real (kind=8), dimension(:), intent(in) distarray_arg,
    integer, dimension(:), intent(in) encounter_indexes_arg,
    integer, intent(in) nb_encounters_arg,
    real(kind=8), intent(in) cutoff_arg,
    character*128, intent(in) output_name_arg )
```

Write selected encounter complexes to an output file.

Writes the header lines from `complexeshead` and then appends encounter lines (from `complexeslines`) in the order provided by `encounter_indexes` until a value exceeding `cutoff` is encountered.

12.12 unit_test/INDEX.md File Reference

12.13 unit_test/QUICK_TEST.md File Reference

12.14 unit_test/README.md File Reference

12.15 unit_test/run_all_tests.sh File Reference

12.16 unit_test/test_assoc.f90 File Reference

Functions/Subroutines

- program [test_assoc](#)
Unit tests for reading association/complexes files.
- subroutine [test_read_assoc_file](#) (num_tests, num_passed, num_failed)
- subroutine [test_allocate_assoc_object](#) (num_tests, num_passed, num_failed)
- subroutine [test_size_assoc](#) (num_tests, num_passed, num_failed)

12.16.1 Function/Subroutine Documentation

12.16.1.1 test_allocate_assoc_object()

```
subroutine test_assoc::test_allocate_assoc_object (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.16.1.2 test_assoc()

program test_assoc

Unit tests for reading association/complexes files.

Comprehensive test suite for reading and parsing association/complexes files in the [read_input](#) and [mod_assoc](#) modules including:

- **File reading:**
 - Valid assoc file reading
 - File existence checking
 - Error handling for missing files
 - Correct number of encounters parsed
- **Data structure allocation:**
 - Allocate assoc objects with correct sizes
 - Initialize translation vectors to zero
 - Initialize rotation vectors (rot1, rot2) to zero
 - Allocate lines array
- **File parsing:**
 - Skip comment lines (starting with #)
 - Count data lines correctly

Author

Abraham Muñoz-Chicharro

Version

1.0

- Extract center coordinates (xc1, xc2)
- Parse translation vectors
- Parse rotation vectors
- **Data integrity:**
 - Verify allocations match encounter count
 - Check header information is stored
 - Verify vector values are reasonable
 - Ensure no data corruption during read

Compile with: gfortran -o test_assoc [test_assoc.f90](#) ../src/read_input.f90 ../src/mod_assoc.f90

Author

Abraham Muñoz-Chicharro

Version

1.0

12.16.1.3 test_read_assoc_file()

```
subroutine test_assoc::test_read_assoc_file (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.16.1.4 test_size_assoc()

```
subroutine test_assoc::test_size_assoc (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.17 unit_test/test_clustering.f90 File Reference

Functions/Subroutines

- program [test_clustering](#)
Unit tests for clustering algorithms in the [mod_clust_algorithm](#) module.
- subroutine [test_clustering_statistics](#) (num_tests, num_passed, num_failed)
- subroutine [test_simple_clustering](#) (num_tests, num_passed, num_failed)
- subroutine [test_minimum_linkage](#) (num_tests, num_passed, num_failed)
- subroutine [test_maximum_linkage](#) (num_tests, num_passed, num_failed)
- subroutine [test_mean_linkage](#) (num_tests, num_passed, num_failed)
- subroutine [test_cluster_writing](#) (num_tests, num_passed, num_failed)

12.17.1 Function/Subroutine Documentation

12.17.1.1 test_cluster_writing()

```
subroutine test_clustering::test_cluster_writing (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.17.1.2 test_clustering()

```
program test_clustering
```

Unit tests for clustering algorithms in the [mod_clust_algorithm](#) module.

Tests include:

- Sort complexes functionality
- Write cluster elements
- Basic clustering threshold calculation
- Minimum linkage clustering
- Maximum linkage clustering
- Mean linkage clustering

Note: Array parameter is mandatory for all clustering methods EXCEPT RMSD. For RMSD-based clustering, array is optional. When provided, array is used to compute cluster averages and standard deviations.

Compile with: gfortran -fopenmp -o test_clustering [test_clustering.f90](#) ../src/mod_clust_algorithm.f90 ../src/read_↵ input.f90

Author

Abraham Muniz-Chicharro

Version

1.0

12.17.1.3 test_clustering_statistics()

```

subroutine test_clustering::test_clustering_statistics (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )

```

12.17.1.4 test_maximum_linkage()

```

subroutine test_clustering::test_maximum_linkage (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )

```

12.17.1.5 test_mean_linkage()

```

subroutine test_clustering::test_mean_linkage (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )

```

12.17.1.6 test_minimum_linkage()

```

subroutine test_clustering::test_minimum_linkage (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )

```

12.17.1.7 test_simple_clustering()

```

subroutine test_clustering::test_simple_clustering (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )

```

12.18 unit_test/test_maths.f90 File Reference**Functions/Subroutines**

- program [test_maths](#)
Unit tests for mathematical operations in the maths module.
- subroutine [test_cross_product](#) (num_tests, num_passed, num_failed)
- subroutine [test_update_complex](#) (num_tests, num_passed, num_failed)
- subroutine [test_vectors_angle_3d](#) (num_tests, num_passed, num_failed)
- subroutine [test_vectors_angle_2d](#) (num_tests, num_passed, num_failed)

- subroutine `test_rmsd` (num_tests, num_passed, num_failed)
- subroutine `test_calculate_cog` (num_tests, num_passed, num_failed)
- subroutine `test_calculate_distance` (num_tests, num_passed, num_failed)
- logical function `assert_vector_equal` (v1, v2, tolerance)

Helper function: Check if two vectors are approximately equal.

12.18.1 Function/Subroutine Documentation

12.18.1.1 `assert_vector_equal()`

```
logical function test_maths::assert_vector_equal (
    real(kind=8), dimension(3), intent(in) v1,
    real(kind=8), dimension(3), intent(in) v2,
    real(kind=8), intent(in) tolerance )
```

Helper function: Check if two vectors are approximately equal.

12.18.1.2 `test_calculate_cog()`

```
subroutine test_maths::test_calculate_cog (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.18.1.3 `test_calculate_distance()`

```
subroutine test_maths::test_calculate_distance (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.18.1.4 `test_cross_product()`

```
subroutine test_maths::test_cross_product (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.18.1.5 `test_maths()`

```
program test_maths
```

Unit tests for mathematical operations in the maths module.

Comprehensive test suite for all subroutines in the maths module including:

- **Cross product:**
 - Basis vectors ($i \times j = k$)
 - Reverse order (anticommutative property)
 - Self cross product ($v \times v = 0$)
 - Arbitrary vectors
- **Coordinate transformations (update_complex):**
 - Identity transformations

- Pure translation
 - Only xc1 translation
 - Only xc2 translation
 - Only rotation (90 degrees)
 - Combined transformations
- **3D angle calculations (vectors_angle_3D):**
 - Parallel vectors (0 degrees)

Author

Abraham Muñoz-Chicharro

Version

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- Perpendicular vectors (90 degrees)
 - Opposite vectors (180 degrees)
 - 45 degree angles
- **2D angle calculations (vectors_angle_2D):**
 - Parallel vectors in XY plane (0 degrees)
 - Perpendicular vectors in XY plane (90 degrees)
 - Opposite vectors in XY plane (180 degrees)
 - Zero magnitude vectors handling
- **RMSD calculation:**
 - Identical coordinates (RMSD = 0)
 - Simple translations
 - Known RMSD values
- **Center of geometry (calculate_cog):**
 - Single point
 - Points at origin
 - Symmetric point distributions
 - General cases
- **Distance calculation (calculate_distance):**
 - Distance to self (0)
 - Unit distances
 - 3D diagonal distances
 - Arbitrary point pairs

Compile with: gfortran -o test_maths [test_maths.f90](#) ../src/maths.f90 ../src/mod_pdb.f90

Author

Abraham Muñoz-Chicharro

Version

1.0

12.18.1.6 test_rmsd()

```
subroutine test_maths::test_rmsd (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.18.1.7 test_update_complex()

```
subroutine test_maths::test_update_complex (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.18.1.8 test_vectors_angle_2d()

```
subroutine test_maths::test_vectors_angle_2d (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.18.1.9 test_vectors_angle_3d()

```
subroutine test_maths::test_vectors_angle_3d (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.19 unit_test/test_matrix.f90 File Reference**Functions/Subroutines**

- program [test_matrix](#)
Unit tests for matrix operations in the [mod_matrix](#) module.
- subroutine [test_write_read_matrix](#) (num_tests, num_passed, num_failed)
- subroutine [test_write_read_array](#) (num_tests, num_passed, num_failed)
- subroutine [test_matrix_z_coord](#) (num_tests, num_passed, num_failed)
- subroutine [test_matrix_rmsd_calc](#) (num_tests, num_passed, num_failed)

12.19.1 Function/Subroutine Documentation**12.19.1.1 test_matrix()**

```
program test_matrix
```

Unit tests for matrix operations in the [mod_matrix](#) module.

Tests include:

- Write and read matrix operations
- Write and read array operations
- Matrix Z-coordinate calculation
- RMSD matrix calculation

Compile with: gfortran -fopenmp -o test_matrix [test_matrix.f90](#) ../src/mod_matrix.f90 ../src/mathsf90 ../src/mod_↵
pdb.f90

Author

Abraham Muñoz-Chicharro

Version

1.0

12.19.1.2 test_matrix_rmsd_calc()

```
subroutine test_matrix::test_matrix_rmsd_calc (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.19.1.3 test_matrix_z_coord()

```
subroutine test_matrix::test_matrix_z_coord (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.19.1.4 test_write_read_array()

```
subroutine test_matrix::test_write_read_array (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.19.1.5 test_write_read_matrix()

```
subroutine test_matrix::test_write_read_matrix (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.20 unit_test/test_pdb.f90 File Reference

Functions/Subroutines

- program [test_pdb](#)
Unit tests for PDB operations in the [mod_pdb](#) module.
- subroutine [test_pdb_allocation](#) (num_tests, num_passed, num_failed)
- subroutine [test_atom_properties](#) (num_tests, num_passed, num_failed)
- subroutine [test_residue_management](#) (num_tests, num_passed, num_failed)

12.20.1 Function/Subroutine Documentation

12.20.1.1 test_atom_properties()

```
subroutine test_pdb::test_atom_properties (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.20.1.2 test_pdb()

program test_pdb

Unit tests for PDB operations in the [mod_pdb](#) module.

Tests include:

- PDB type allocation and initialization
- Atom and residue management
- Chain structure handling

Compile with: gfortran -o test_pdb [test_pdb.f90](#) [mod_pdb.f90](#)

Author

Abraham Muñiz-Chicharro

Version

1.0

12.20.1.3 test_pdb_allocation()

```
subroutine test_pdb::test_pdb_allocation (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.20.1.4 test_residue_management()

```
subroutine test_pdb::test_residue_management (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.21 unit_test/test_read_input.f90 File Reference

Functions/Subroutines

- program [test_read_input](#)
Unit tests for the [read_input](#) module.
- subroutine [test_read_input_assoc](#) (num_tests, num_passed, num_failed)
- subroutine [test_error_handling](#) (num_tests, num_passed, num_failed)

12.21.1 Function/Subroutine Documentation

12.21.1.1 test_error_handling()

```
subroutine test_read_input::test_error_handling (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.21.1.2 test_read_input()

```
program test_read_input
```

Unit tests for the [read_input](#) module.

Comprehensive test suite for reading PDB and association/complexes files using the [read_input](#) module wrapper functions including:

- **PDB file reading (read_pdb):**
 - Valid PDB file reading
 - File existence checking
 - Atom count verification
 - Residue count verification
 - Chain count verification
 - Proper structure allocation
 - Error handling for missing files
- **Association file reading (read_assoc):**
 - Valid assoc file reading
 - Encounter count verification
 - Structure allocation and initialization
 - Vector data integrity
 - Error handling for missing files

Author

Abraham Muñoz-Chicharro

Version

1.0

- **Integration tests:**
 - Sequential reading of multiple files
 - Proper cleanup between reads
 - Memory management

Compile with: gfortran -o test_read_input [test_read_input.f90](#) ../src/read_input.f90 ../src/mod_pdb.f90 ../src/mod_↵
_assoc.f90

Author

Abraham Muñoz-Chicharro

Version

1.0

12.21.1.3 test_read_input_assoc()

```
subroutine test_read_input::test_read_input_assoc (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.22 unit_test/TEST_SUMMARY.md File Reference**12.23 unit_test/test_threshold.f90 File Reference****Functions/Subroutines**

- program [test_threshold](#)
Unit tests for threshold operations in the [mod_threshold](#) module.
- subroutine [test_array_z_coord](#) (num_tests, num_passed, num_failed)
- subroutine [test_array_atoms_dist](#) (num_tests, num_passed, num_failed)
- subroutine [test_array_angle](#) (num_tests, num_passed, num_failed)
- subroutine [test_sort_array](#) (num_tests, num_passed, num_failed)

12.23.1 Function/Subroutine Documentation**12.23.1.1 test_array_angle()**

```
subroutine test_threshold::test_array_angle (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.23.1.2 test_array_atoms_dist()

```
subroutine test_threshold::test_array_atoms_dist (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.23.1.3 test_array_z_coord()

```
subroutine test_threshold::test_array_z_coord (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.23.1.4 test_sort_array()

```
subroutine test_threshold::test_sort_array (
    integer, intent(inout) num_tests,
    integer, intent(inout) num_passed,
    integer, intent(inout) num_failed )
```

12.23.1.5 test_threshold()

program test_threshold

Unit tests for threshold operations in the [mod_threshold](#) module.

Tests include:

- Z-coordinate array calculation
- Minimum distance calculation between atoms
- Angle computation for different dimensions
- Array sorting with encounter index tracking

Compile with: gfortran -fopenmp -o test_threshold [test_threshold.f90](#) ../src/mod_threshold.f90 ../src/mathsf90
../src/mod_pdb.f90

Author

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Version

1.0

12.24 unit_test/TESTING.md File Reference

12.25 unit_test/UNIT_TESTING_COMPLETE.md File Reference

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